Overview of windows central management

CYBEXER TECHNOLOGIES

Microsoft Intune

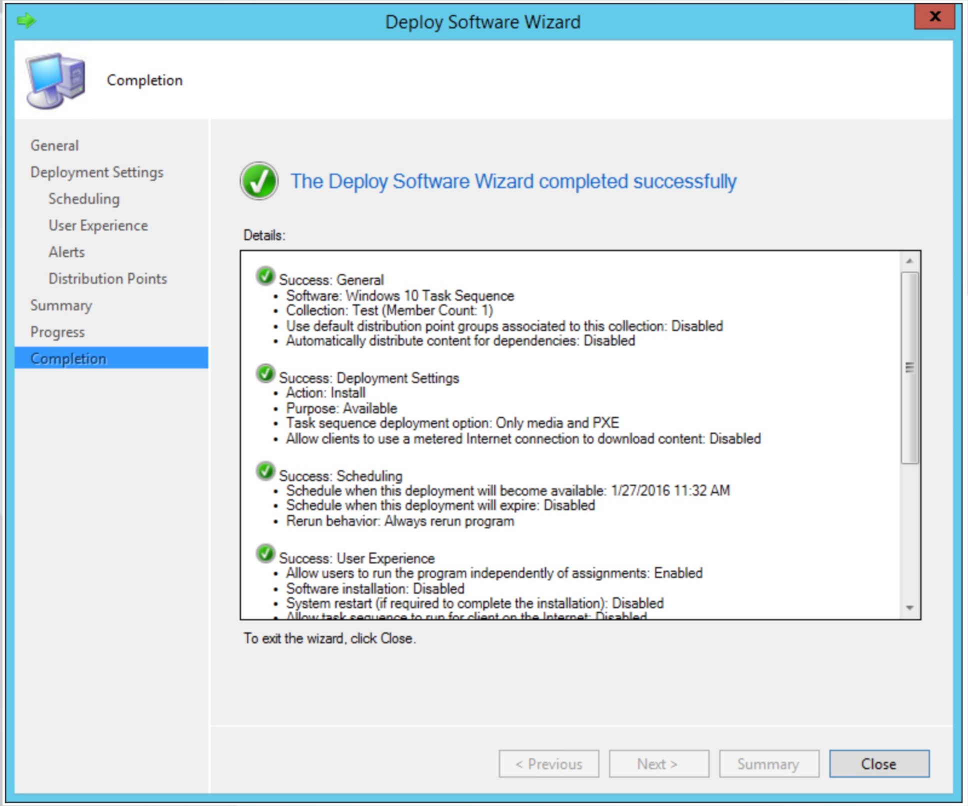
- Web based
- Focused heavily on mobile device management
- Sold as a cloud service on a subscription basis
- Very strong focus on corporate data and data leak protection (DLP)
  - Classify documents, emails, pictures
  - Prevent access to documents,emails,pictures on an unprotected device
  - Watermark documents / pictures

Microsoft SCCM

- Focuses more on workstations and servers
- Main features are
  - Configuration management
  - Patch management
  - Software distribution
  - Operating system deployment
  - Hardware and software inventory
  - Power management
- PXE boot and bare metal imaging

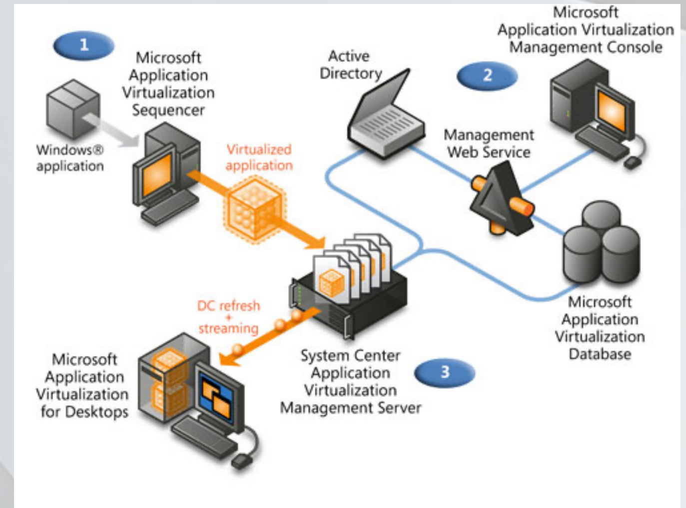
SCCM Task sequencer

- One of the biggest advantages of SCCM is to create automated process to prepare workstations / servers



Application Virtualization (App-V)

- Streaming application instead of installing it
- Whole application lives inside a ”bubble” and is streamed from an AppV server
- OS tampering is minimal due to the missing of “install” or “uninstall”



ACTIVE DIRECTORY

Logical components of AD DS

- Partition
- Schema
- Domain
- Domain tree
- Forest
- OU
- Container

Partition

- A partition, or naming context, is a portion of the AD DS database. Although the database consists of one file named Ntds.dit, different partitions contain different data. For example, the schema partition contains a copy of the Active Directory schema. The configuration partition contains the configuration objects for the forest, and the domain partition contains the users, computers, groups, and other objects specific to the domain. Active Directory stores copies of partitions on multiple domain controllers and updates them through directory replication.

Schema

- A schema is the set of definitions of the object types and attributes that you use to define the objects created in AD DS.

Domain

- A domain is a logical administrative container for objects such as users and computers. A domain maps to a specific partition and you can organize the domain with parent-child relationships to other domains.

Domain tree

- A domain tree is a hierarchical collection of domains that share a common root domain and a contiguous Domain Name System (DNS) namespace.

Forest

- A forest is a collection of one or more domains that have a common AD DS root, a common schema, and a common global catalog.

Organizational Unit (OU)

- An OU is a container object for users, groups, and computers that provides a framework for delegating administrative rights and administration by linking Group Policy Objects (GPOs).

Container

- A container is an object that provides an organizational framework for use in AD DS. You can use the default containers, or you can create custom containers. You can't link GPOs to containers.

Physical components of AD DS

- Domain controller
- Data store
- Global catalog server
- Read-only domain controller (RODC)
- Site
- Subnet

Domain controller

- A domain controller contains a copy of the AD DS database. For most operations, each domain controller can process changes and replicate the changes to all the other domain controllers in the domain.

Data store

- A copy of the data store exists on each domain controller. The AD DS database uses Microsoft Jet database technology and stores the directory information in the Ntds.dit file and associated log files. The C:\Windows\NTDS folder stores these files by default.

Global catalog server

- A global catalog server is a domain controller that hosts the global catalog, which is a partial, read-only copy of all the objects in a multiple-domain forest. A global catalog speeds up searches for objects that might be stored on domain controllers in a different domain in the forest.

Read-only domain controller (RODC)

- An RODC is a special, read only installation of AD DS. RODCs are common in branch offices where physical security is not optimal, IT support is less advanced than in the main corporate centers, or line-of-business applications need to run on a domain controller.

Site

- A site is a container for AD DS objects, such as computers and services that are specific to a physical location. This is in comparison to a domain, which represents the logical structure of objects, such as users and groups, in addition to computers.

Subnet

- A subnet is a portion of the network IP addresses of an organization assigned to computers in a site. A site can have more than one subnet.

User objects in AD DS

- User account enables access to network resources where the user can authenticate to AD DS to access the resources.
- User account includes
  - username
  - user password
  - group memberships
  - many other organizational information fields

Managed service accounts (MSA)

- Many applications contain services that are run in background and which enable interaction with the application
- Often the applications don’t need any user interaction but might require access to additional resources in the AD DS (network shares, files and folders on the serving computer)
- MSA can be local to the computer (Local service, Network Service, Local System) or domain based

Domain based account vs MSA

- Disadvantages of domain based account
  - Extra administrative effort might be necessary to manage account password (securely)
  - Might be hard to identify if a domain-based account is being used as a service account
  - Extra administration effort might be necessary to manage service principal name (SPN)
- Benefits of MSA
  - Simplified password management
  - Simplified SPN management

Group based managed service accounts

- Group managed service accounts enable the use of MSA in more than one server in the domain (ie load balancing clusters or IIS servers)
- To support group managed service accounts (gMSA) a KDS (Key Distribution Services) root key needs to be created on a domain controller
  - Add-KdsRootKey -EffectiveImmidiately
  - New-ADServiceAccount -Name SQLfarm –PrincipalsAllowedToRetrieveManagedPassword SERVER1,SERVER2,SERVER3

Group objects

- Groups allow permission management over set of objects (users,computers)
- Group can be either
  - Security group - security enabled group to assign permissions to various resources. Security groups can be used to manage access control lists (ACLs)
  - Distribution - email applications use distribution groups which are not security-enabled. Security groups can be used as distribution groups for email applications

Group scope

- Group scope determines the range of abilities or the permissions and the group membership
- 4 types of scopes are supported
  - Local
  - Domain-local
  - Global
  - Universal

Group scope - local

- Local - one can use this type of group for standalone servers or workstations, on domain-member servers that are not domain controllers or on domain-member workstations. Local groups are available only on the computer where they exist. Important characteristics are
  - One can only assign abilities and permissions to local resources (on local computer)
  - Members can be from anywhere in the AD DS forest

Group scope domain-local

- This type of group is primary meant to manage access to resources or assign management rights and responsibilities. Domain-local groups exist on domain controllers and the groups scope is located to the domain in which it resides. Important characteristics are
  - Abilities and permissions can be assigned to domain-local resources only which means on all the computers in the local domain
  - Members can be from anywhere in the forest

Group scope - global

- This group can be used to consolidate users who have similar characteristics – for example to reference users who are part of a department or a geographic location. The characteristics of the global groups are
  - One can assign abilities and permissions anywhere in the forest
  - Members can be from the local domain only and can include users, computers and global groups from the local domain

Group scope - universal

- Universal scope is used mostly in multidomain networks because it combines the characteristics of domain-local and global groups. The important characteristics are
  - Abilities and permissions can be assigned anywhere in the forest similar to global groups
  - Members can be from anywhere in the AD DS forest

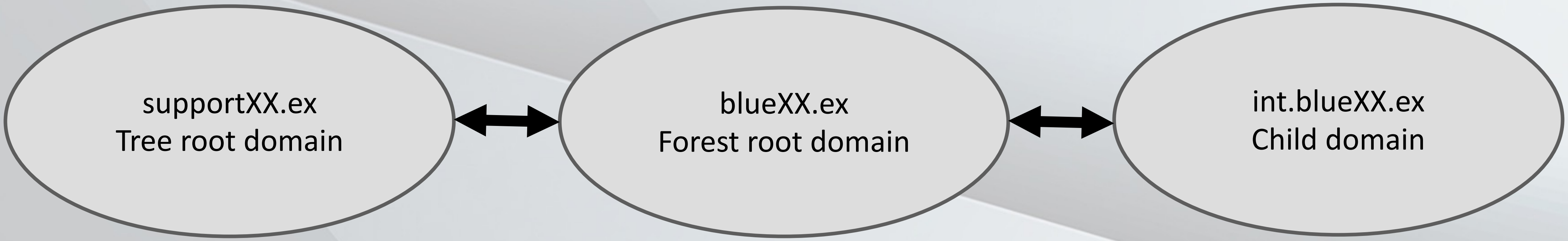
Computer objects

- Computers are similar to users as they
  - Have an account with sign in name and password that windows rotates automatically
  - Authenticate to the domain
  - Can belong to groups and have access to resources and can be configured by using group policy
- Computers container
  - Built-in container where by default all the domain joined computers are added
  - It differs from an OU as one can not apply group policies to a container

AD DS forest

- A forest is a top level container in AD DS
- Each forest is a collection of one or more domain trees that share common directory schema and a global catalog.
- A domain tree is a collection of one or more domains that share a contiguous namespace
- The forest root domain is the first domain that will be created when a new forest is created
- Forest root contains objects that do not exist in other domains in the forest as those objects are created with the first domain controller
  - Enterprise admins
  - Schema admins
- A forest can consist of minimum one domain controller to a limit of several domains across multiple domain trees

AD DS forest


supportXX.ex
Tree root domain
blueXX.ex
Forest root domain
int.blueXX.ex
Child domain

Important characteristics of AD DS forest

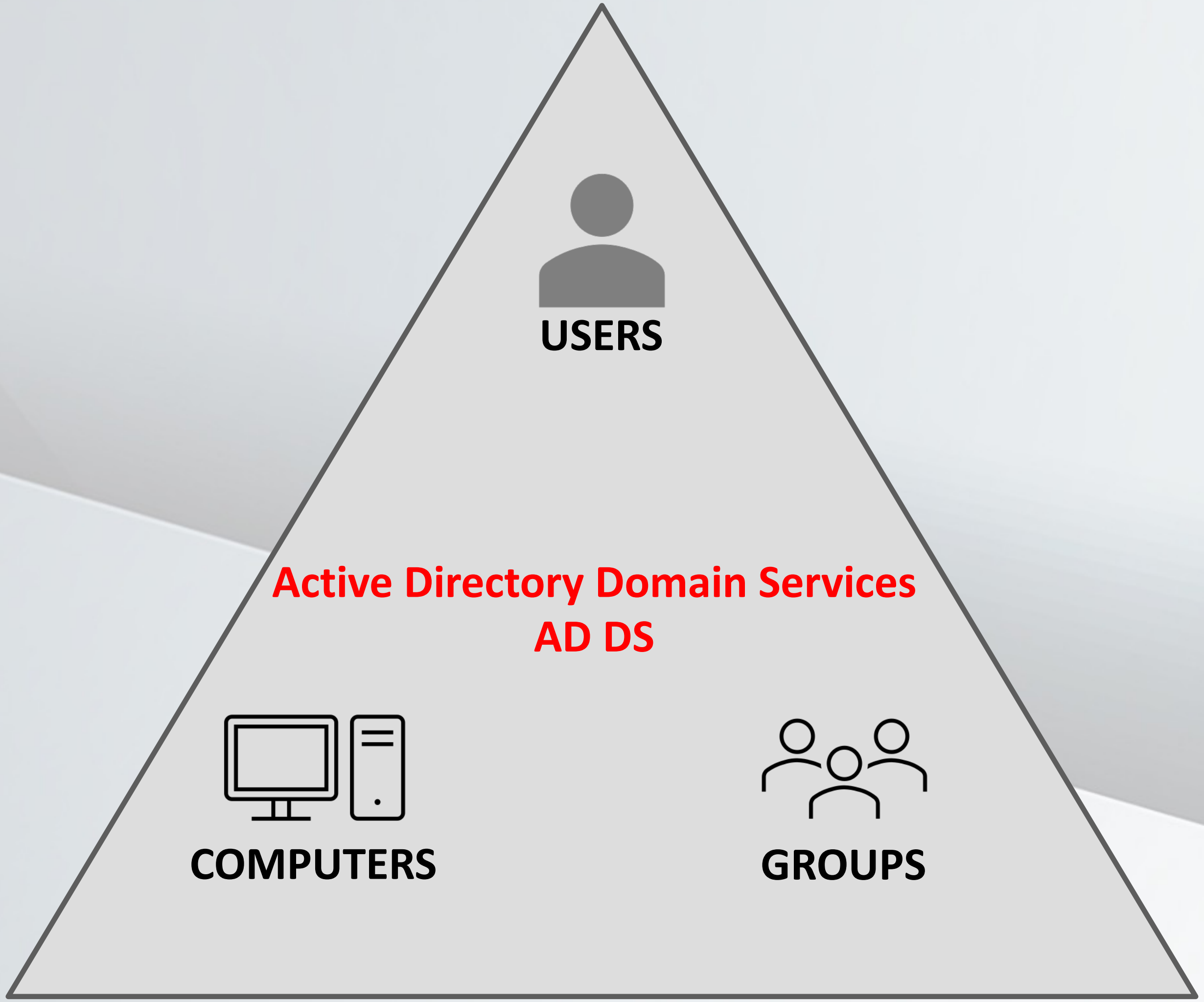
- A security boundary. By default, no users from outside the forest can access any resources inside the forest. In addition, all the domains in a forest automatically trust the other domains in the forest. This makes it easy to enable access to resources for all the users in a forest, regardless of the domain to which they belong.
- A replication boundary. An AD DS forest is the replication boundary for the configuration and schema partitions in the AD DS database. Therefore, organizations that want to deploy applications with incompatible schemas must deploy additional forests. The forest is also the replication boundary for the global catalog. The global catalog makes it possible to find objects from any domain in the forest.

Objects in forest root domain

- Schema master
  - Created with the first domain controller but can be moved to other domain controllers in the forest
- Domain naming master
  - Created with the first domain controller but can be moved to other domain controllers in the forest
- Enterprise Admins group
- Schema admins group

AD DS domain

- User accounts
- Computer accounts
- Groups



Objects in each domain including forest root

- RID master role
- Infrastructure master role
- PDC emulator master role
- Domain Admins group

AD DS key benefits

- Authentication. Whenever a domain-joined computer starts or a user signs in to a domain-joined computer, AD DS authenticates it. Authentication verifies that computer or user has the proper identity in AD DS by verifying its credentials.
- Authorization. Windows uses authorization and access control technologies to determine whether to allow authenticated users to access resources.

Trusts in AD DS

- Trusts enable access to resources in complex AD DS environment
- In multi domain AD DS forest two-way transitive trust relationships generate automatically between AD DS domains so that a path of trust exists between all AD DS domains

Trust types

Trust type	Description	Direction	Description
Parent and child	Transitive	Two-way	When you add a new AD DS domain to an existing AD DS tree, you create new parent and child trusts.
Tree-root	Transitive	Two-way	When you create a new AD DS tree in an existing AD DS forest, you automatically create a new tree-root trust.
External	Nontransitive	One-way or two-way	External trusts enable resource access with a Windows NT 4.0 domain or an AD DS domain in another forest. You also can set these up to provide a framework for a migration.
Realm	Transitive or nontransitive	One-way or two-way	Realm trusts establish an authentication path between a Windows Server AD DS domain and a Kerberos version 5 (v5) protocol realm that implements by using a directory service other than AD DS.
Forest (complete or selective)	Transitive	One-way or two-way	Trusts between AD DS forests allow two forests to share resources.
Shortcut	Nontransitive	One-way or two-way	Configure shortcut trusts to reduce the time taken to authenticate between AD DS domains that are in different parts of an AD DS forest. No shortcut trusts exist by default, and an administrator must create them if they are required.

Organizational units (OUs)

- An OU is a container object within a domain that you can use to consolidate users, computers, groups, and other objects. You can link Group Policy Objects (GPOs) directly to an OU to manage the users and computers contained in the OU. You can also assign an OU manager and associate a COM+ partition with an OU.

Where to use OUs

- Two main reasons to create OU
  - To consolidate objects to make it easier to manage them by applying GPOs to the collective. When you assign GPOs to an OU, the settings apply to all the objects within the OU. GPOs are policies that administrators create to manage and configure settings for computers or users. You deploy the GPOs by linking them to OUs, domains, or sites.
  - To delegate administrative control of objects within the OU. You can assign management permissions on an OU, thereby delegating control of that OU to a user or a group within AD DS, in addition to the Domain Admins group.
- One can use OUs to represent the hierarchical, logical structures within your organization. For example, you can create OUs that represent the departments within your organization, the geographic regions within your organization, or a combination of both departmental and geographic regions. You can use OUs to manage the configuration and use of user, group, and computer accounts based on your organizational model

Containers

- AD DS has several built-in containers, or generic containers, such as Users and Computers. These containers store system objects or function as the default parent objects to new objects that you create.
- One should not confuse these generic container objects with OUs. The primary difference between OUs and containers is the management capabilities. **Containers have limited management capabilities. For example, you can't apply a GPO directly to a container.**

Objects created during AD DS creation

- Domain. The top level of the domain organizational hierarchy.
- Builtin container. A container that stores several default groups.
- Computers container. The default location for new computer accounts that you create in the domain.
- Foreign Security Principals container. The default location for trusted objects from domains outside the local AD DS domain that you add to a group in the local AD DS domain.
- Managed Service Accounts container. The default location for managed service accounts. AD DS provides automatic password management in managed service accounts.
- Users container. The default location for new user accounts and groups that you create in the domain. The Users container also holds the administrator, the guest accounts for the domain, and some default groups.
- Domain Controllers OU. The default location for domain controllers' computer accounts. This is the only OU that is present in a new installation of AD DS.

Containers visible in advanced features

- LostAndFound - This container holds orphaned objects.
- Program Data - This container holds Active Directory data for Microsoft applications, such as Active Directory Federation Services (AD FS).
- System - This container holds the built-in system settings.
- NTDS Quotas - This container holds directory service quota data.
- TPM Devices - This container stores the recovery information for Trusted Platform Module (TPM) devices.

Best practices setting up OU structure

- The administrative needs of the organization dictate the design of an OU hierarchy. Geographic, functional, resource, or user classifications could all influence the design. Whatever the order, the hierarchy should make it possible to administer AD DS resources as effectively and flexibly as possible. For example, if you need to configure all IT administrators' computers in a certain way, you can group all the computers in an OU and then assign a GPO to manage those computers.
- One can create OUs within other OUs. For example, your organization might have multiple offices, each with its own IT administrator who is responsible for managing user and computer accounts. In addition, each office might have different departments with different computer-configuration requirements. In this situation, you can create an OU for each office, and then within each of those OUs, create an OU for the IT administrators and an OU for each of the other departments.
- Although there is no limit to the number of levels in your OU structure, **limit your OU structure to a depth of no more than 10 levels to ensure manageability.** Most organizations use five levels or fewer to simplify administration. Note that applications that work with AD DS can impose restrictions on the OU depth within the hierarchy for the parts of the hierarchy that they use.

AD DS MANAGEMENT AND FSMO ROLES

Restoring AD DS deleted objects by using recycle bin

- Windows server offers Active Directory Recycle Bin feature to enable deleted object restoration without OS downtime
- When objects are deleted from AD they are actually moved to Active Directory Recycle Bin and remain there until the deleted object lifetime expires (by default 180 days in new AD DS deployments)

AD DS backup and restore

- Backup must include system state data which includes collection of critical OS and server role files in addition to AD DS database and registry
- To perform restore one must have full acces to the files on domain controller which means the DC has to be restarted into DSRM (Directory Services Restore Mode).
- Nonauthoritative restore
  – By default AD DS is restored to a known good state and all subsequent updates to the database are replicated from other domain controllers. Typically used when the domain controller / domain controller data has been damaged or corrupted but the issues have not replicated to other domain controllers
- Authoritative restore
  – Once a authoritative restore has been performed the data is marked as authoritative and replicated outbound to all replication partners (other DC-s)

AD DS global catalog role

- Global catalog is partial, read-only, searchable copy of all the objects in a forest.
- Main objective is to speed up searches for objects that might be stored on domain controllers in different domain in the forest
- In single domain AD DS database on each domain controller contains all the information about every object in that domain. In a search query only known information is returned (specific to the domain)
- To include information from all forest query must be made to a domain controller which is also a global catalog
- Global catalog DOES NOT INCLUDE ALL ATTRIBUTES FOR EACH OBJECT ! Instead it maintains a subset of attributes that are most likely to be useful in cross-domain searches (for example mail,displayName etc).
- The attributes replicated to the GC are defined in schema and can be altered

AD DS operations masters

- AD DS operation masters roles are responsible for performing operations that are not suitable for a multiple-master model
- Operations master role is known as a Flexible Single Master Operation (FSMO) role
  - Schema master
  - Domain-naming master
  - Infrastructure master
  - Relative ID (RID) master
  - Primary domain controller (PDC) emulator master
- By default the first domain controller installed in a forest hosts all 5 roles
- Roles can be transferred later
- Each forest has one schema master and one domain naming master
- Each AD DS domain has one RID master, one infrastructure master and one PDC emulator

Forest operational masters

- Domain naming master
  - Responsible when domains are added or removed in forest
  - If domain naming master is unavailable the changes cannot be made
- Schema master
  - Schema master is the domain controller that will make all schema changes
  - If schema master is unavailable the changes to the domain schema CANNOT be made

Domain operations masters

- RID master
  - When new objects are created in AD DS they will get unique security ID (SID)
  - To ensure that no two or more domain controllers assign the same SID for different objects the RID master allocates blocks of RIDs to domain controllers from where they can assign SID-s
  - If RID master is unavailable there might be issues adding security principals like users,computers etc to the domain as if the domain controller has exhausted the given block of SID-s it is unable to create new objects
- Infrastructure master
  - This role maintains interdomain object references ie when a group in domain A has members from domain B then infrastructure master helps to translate the used SID-s into names
  - If infrastructure master is unavailable then domain controllers who are not global catalogs will not be able to translate SIDs into security principal names
  - Infrastructure master should preferably not reside on a domain controller that is hosting the global catalog unless every domain controller in the forest is configured to serve GC- in this case the infrastructure master role is actually unnecessary because DC-s know about all objects

Domain operations masters

- PDC emulator master
  - PDC emulator master serves as the time source for the domain
  - PDC master in each domain synchronizes the time with the PDC master in forest root domain
  - PDC master time in forest root domain should be synced with reliable external time source
  - By default changes to the GPO-s are written to the PDC emulator master
  - If a user password is changed then it is immediately communicated to the PDC master and when a user tries to log in then PDC emulator is queried for the most recent changes
  - If PDC emulator master is unavailable users might have trouble signing in until their password changes have replicated to all the domain controllers

Managing AD DS operations masters

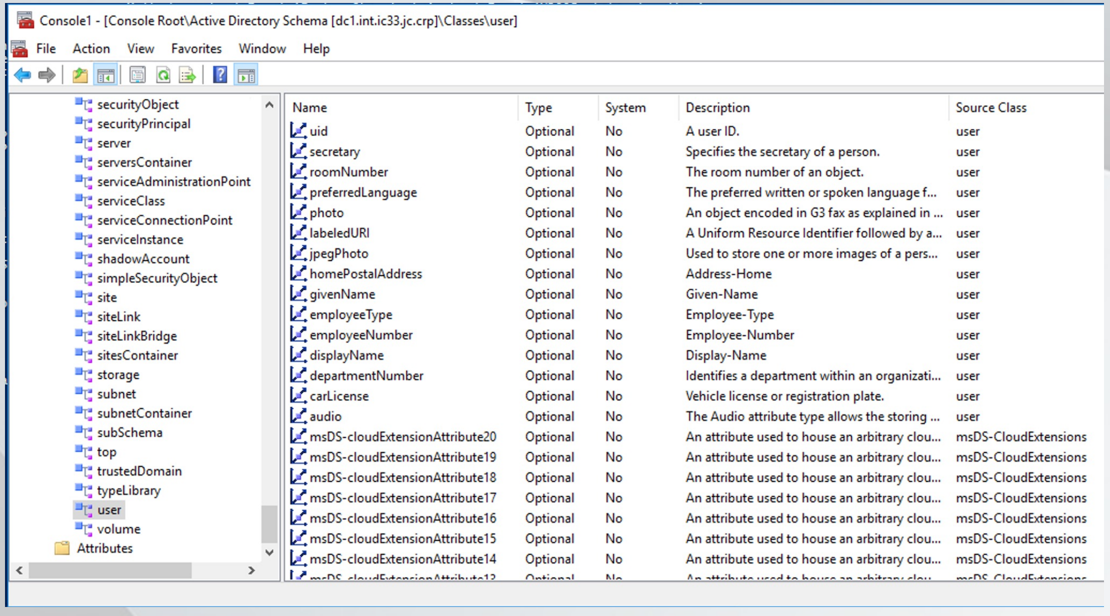
- Operations masters should be distributed in the forest according to the respecting roles, network latency, users etc
- FSMO roles can be transferred (preferred way) and seized (last resort)
  - **Move-ADDirectoryServerOperationsMasterRole –Identity "<servername>" -OperationsMasterRole "<rolenamelist>" -Force**
    - The -Force indicates seizing
  - netdom query fsmo
    - To query who holds which role
  - ntdsutil can also be used to transfer roles but is not a oneliner option
    - first type “roles” then type “connections” then “connect to your.server.full.name” and then type “q”
    - Then type one of the following commands to transfer the roles
Transfer domain naming master
Transfer infrastructure master
Transfer PDC
Transfer RID master
Transfer schema master
    - After transfer is done type “q” to quit

AD DS schema

- As active directory is just a multi master database then all the objects in there are stored according to the described schema
- Schema defines all object types
- Object definitions specify the
  - type of the data to store
  - data syntax
- Only objects defined by schema can be created
- AD DS schema objects consist of attributes which are grouped together into classes
- Class rule defines which attributes are mandatory and which are optional – for example the user class can consist of more than 400 different attributes including the common name (cn), givenName,objectSID,uid,gid (for linux userid and groupid)

AD DS schema

- User class is an example of structural class – structural class is the only type of class that can have objects in an AD DS database
- To modify schema an auxiliary class with attributes can be created and then reference it in the definition of a structural class
- To enable schema management in mmc.exe first register the schema dll
  regsvr32 schmmgmt.dll



MICROSOFT GROUP POLICY

Group Policy

- What is Group Policy ?
  - A centralized management and configuration tool for computers and users. No extra client or server beyond a Domain Controller (DC) is required
- What can it do ?
  - Pretty much everything- endless amounts of on / off settings or value changes, deploy scripts and manage access
- What is it comprised of ?
  - A service (GPClient) on every machine, Active Directory / Sysvol for group policy objects (GPOs) a group policy management console (GPMC)

Definitions related to Group Policy

- Group Policy: the overall technology. Named because you are grouping policies and settings together.
- Group Policy Object (GPO): an individual object that contains your settings
- Policy (setting): also known as a registry setting or administrative setting. These are the individual Enable/Disable settings you can configure in a GPO
- policy: a specific type of setting that users cannot change
- Group Policy Preferences (GPP): an add-on for Group Policy that is built into Windows 7. An example would be Group Policy Preferences (GPP): Printers
- Preference (setting): an individual item within a specific GPP (like Printers)
- preference: a specific type of setting that users can change

Tattooed vs non-tattooed settings

- Tattooing is the effect that whenever you apply a registry policy to a computer or user and then remove that policy file then even though the policy is gone, those registry values that were set by the policy remained–tattooed into the registry until you explicitly remove them, either by setting the policy to the opposite value or manually going in and deleting the registry values
- Non-tattooed settings will be removed once the relevant group policy setting is removed

Where do the settings live?

- When you configure an individual policy (ex: Remove Change Password", it is stored on clients in a special location.

- If it is in the computer node:
  - HKLM\Software\Policies
  - HKLM\Software\Microsoft\Windows\CurrentVersion\Policies

- If it is in the user node:
    - HKCU\Software\Policies
  - HKCU\Software\Microsoft\Windows\CurrentVersion\Policies

What is a preference?

- Any setting that tattoos a setting – as in leaves the setting behind when no longer applicable
- Any setting that allows the user to work around it.

- Biggest example is Group Policy Preferences.

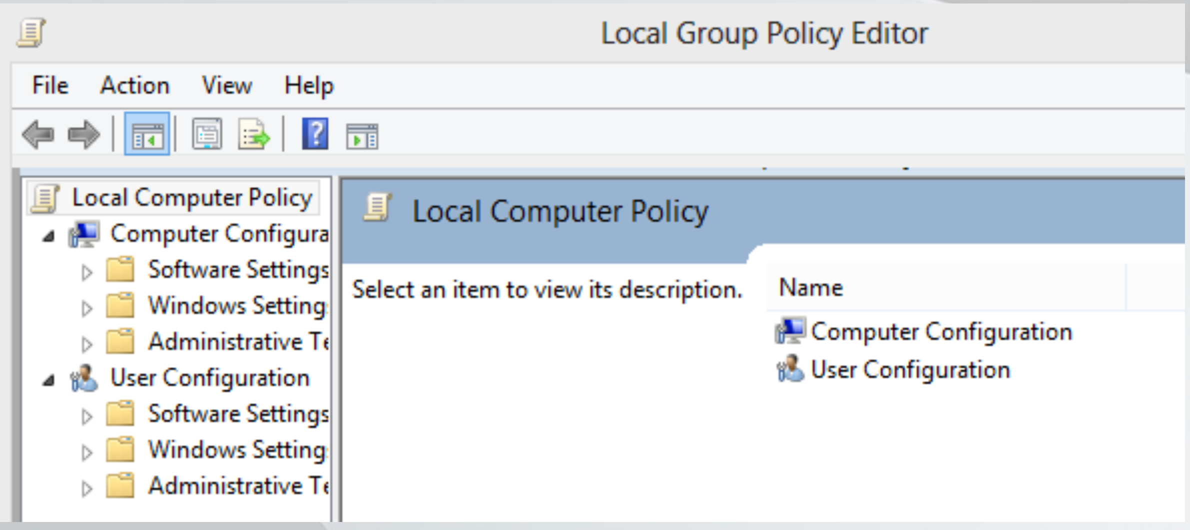
- Another example: Internet Explorer Maintenance (yes, the same one that lives in the Policies node)
  - Stores settings at: HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings

Why use a preference?

- To set default settings.
  - Ex: You would like to set the default homepage but allow users to change it
- To configure applications that don’t support policy
  - Using GPPrefs: Files, Registry, INI, etc to deploy preconfigured settings
- To eliminate logon scripts
  - GPPrefs are faster, GUI based, and highly configurable

Local Group Policy Objects (LGPOs)

- Business Editions of Windows include at least 1 local Group Policy Object
- Applies to all users logging into a machine
- Accessed by Launching GPEDIT.MSC
- Stored at: C:\windows\system32\grouppolicy



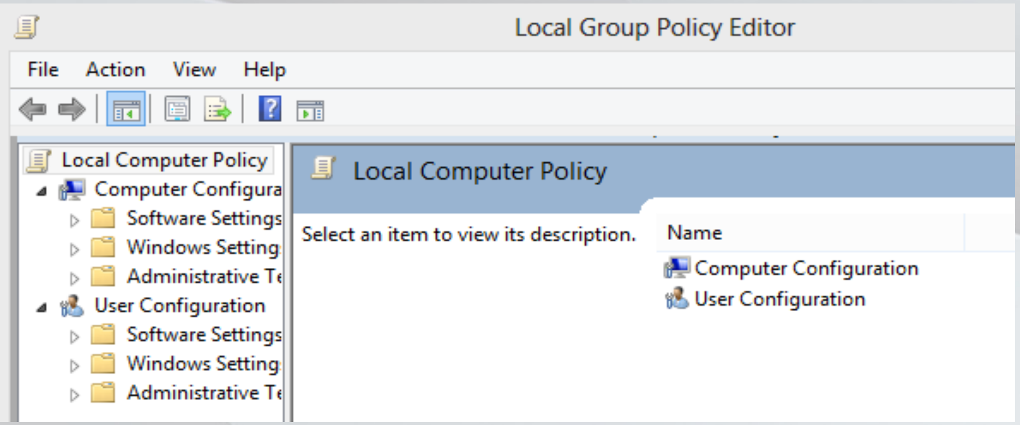
LGPOs VS Active Directory GPOs

- LGPOs
  - Work in off domain environments
  - Extremely fast to process
  - Have to be set on a machine by machine basis
  - Not all features available such as Software Installation

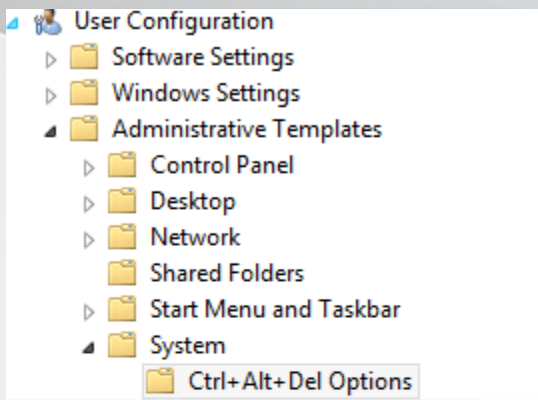
- AD GPOs
  - For domains only
  - Slightly slower to process
  - Able to massively set changes
  - Tons of features and extensions

Users vs Computers

GPOs are divided up into two sections: Computer and User. These are collectively called nodes.

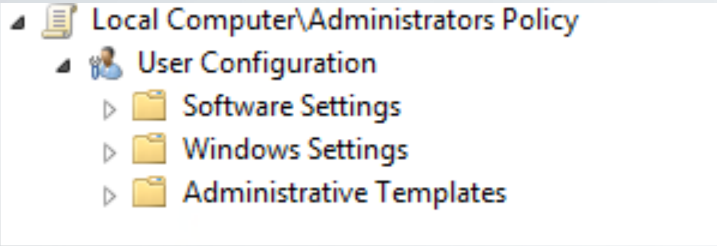


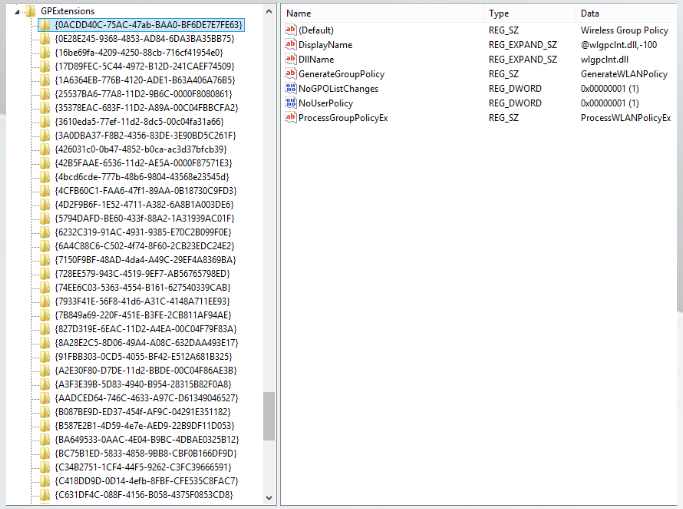
- So if said, find User Configuration/Administrative Templates/System – here is where you would look:



Pieces of a Policy

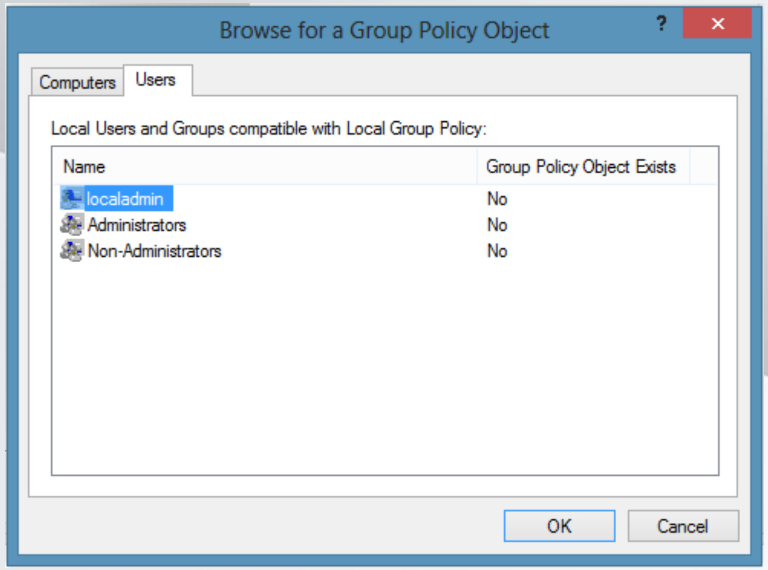
- A single GPO lets you configure options that are processed by many Client Side Extensions (CSEs)
- CSEs add functionality to GP
  - Software Installation
  - Security
  - Registry
  - Each Preference: Printers, Files
- Full list: HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon\GPExtensions



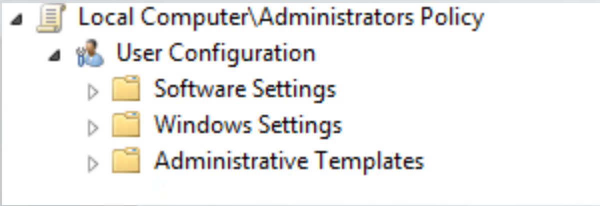


Multiple Local GPOs

- Starting in Vista, Local GPOs can be applied to different user types:
  - The Computer Only – settings here apply to all users
  - Administrators Only
  - Non-Administrators Only
  - A specific user

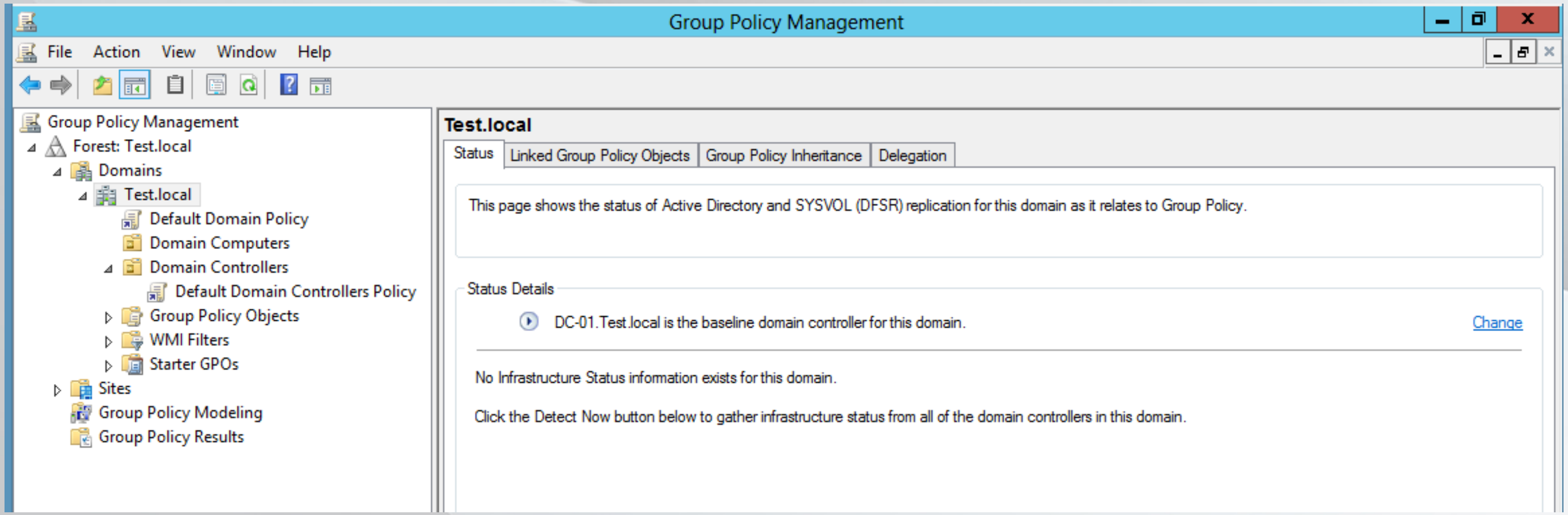


- Accessed by launching MMC Loading Group Policy Object Editor and browsing Users tab.



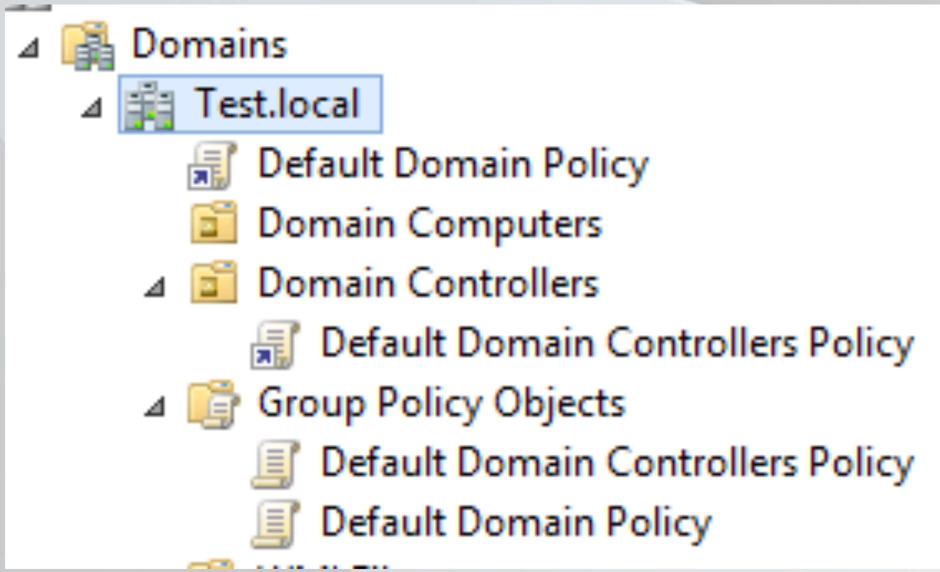
Group Policy Management Console (GPMC)

- GPEdit allows for management of single GPO
- GPMC let’s you see the whole forest – GPOs, links, results, replication, etc



Creating or Linking

- GPOs live in the Group Policy Objects Container.
- In ADUC, you can find them under the Policies Container
- GPOs are linked to an OU. Basically, a link is a shortcut. One GPO can be linked to many OUs

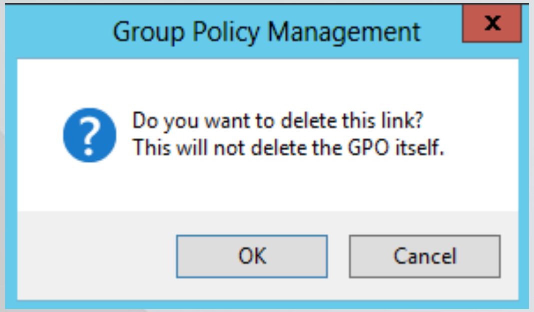


GP Editing

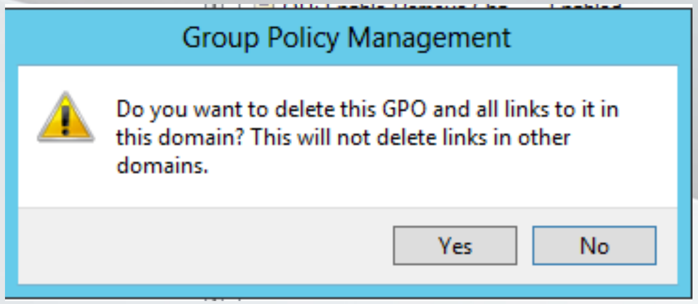
- You can edit a GPO by right clicking on a link or right clicking on the actual GPO itself.
- Any editing is live! If a GPO is linked, clients can start picking up your edits before you are finished
- Editing a GPO that is linked to multiple OUs means that every one of those OUs will get your changes

GP Deleting

- Because GPOs live in the Group Policy Objects container, they can only be deleted there.
- This is the message when you Delete a link to an OU:

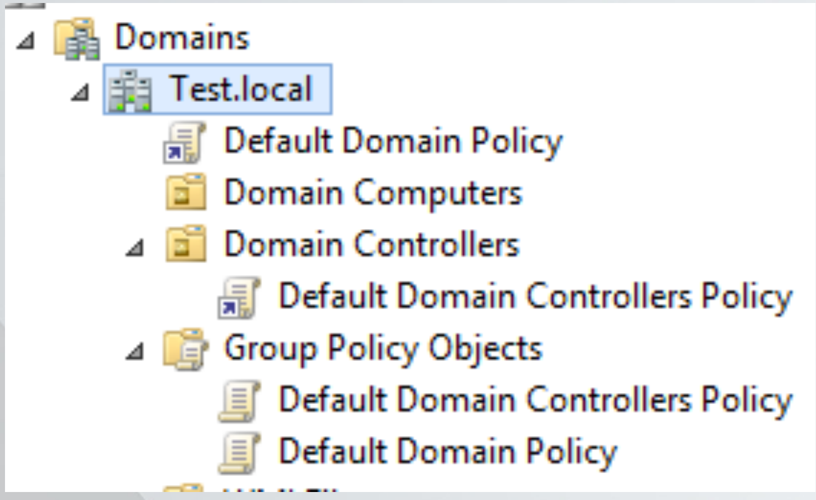


- This is the message you'll get when deleting a GPO.



Built-In GPOs

- Two GPOs exist by default
  - Default Domain
  - Default Domain Controller
- Do not edit these (unless to change password policy) – separate changes in separate GPOs
- Why? Performance and Standardization



Safety net in case Default policies have been changed

- To revert back to the default settings.
  - Compare your GPOs to the default GPOs shipped
  - Set those settings in a dedicated policy
  - Use DCGPOFix to revert back

```
Syntax: DcGPOFix [/ignoreschema] [/Target: Domain | DC | BOTH]

This utility can restore either or both the Default Domain Policy or the
Default Domain Controllers Policy to the state that exists immediately after
domain creation. You must be a domain administrator to perform this operation.

WARNING: YOU WILL LOSE ANY CHANGES YOU HAVE MADE TO THESE GPOs. THIS UTILITY
IS INTENDED ONLY FOR DISASTER RECOVERY PURPOSES.

You are about to restore Default Domain Policy and Default Domain Controller Po
icy for the following domain
Test.local
Do you want to continue: <Y/N>?
```

How Group Policy Updates

- Two types of updates: foreground and background
  - Foreground: a computer starts, a user logs on, a system is brought out of hibernation, and at network changes
  - Background:
    - Clients: every 90 minutes + 30 minute salt. Max is 120 minutes.
    - Domain Controllers: every 5 minutes + No Salt. For large organizations, you can add random delay

- Two types of methods: Synchronous and Asynchronous
  - Synchronous: CSEs process in order – when one finishes, the next begins. User logs in when all have finished.
  - Asynchronous: CSEs process multi threaded – not dependent on each other. User logs in while still processing.

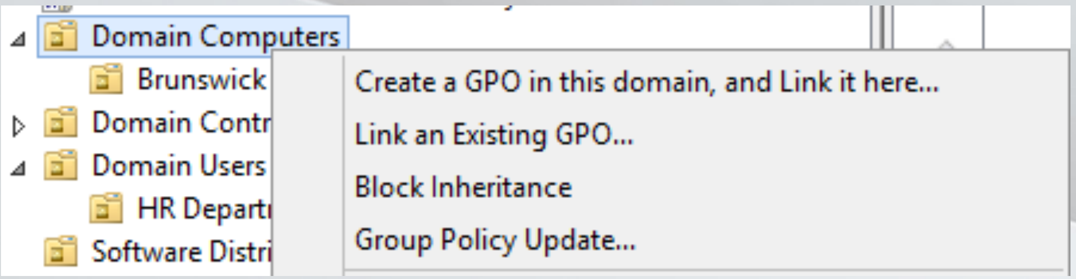
- Some CSEs are always foreground and synchronous.
  - Drive Mappings, Folder Redirection, Software Installation

GPUpdate

- GPUpdate.exe is the tool used to trigger a Group Policy Update
- When an update is triggered, only new policies are applied.
- To run, just type GPUpdate from CMD. Only GPOs that have changed will be processed.
- To reprocess all GPOs, add a /force

Remote GPUpdate

- GPMC for Windows 8 – right click OU and select Group Policy Update
- How does it work ? Firewall ?



Configuring Update Behavior

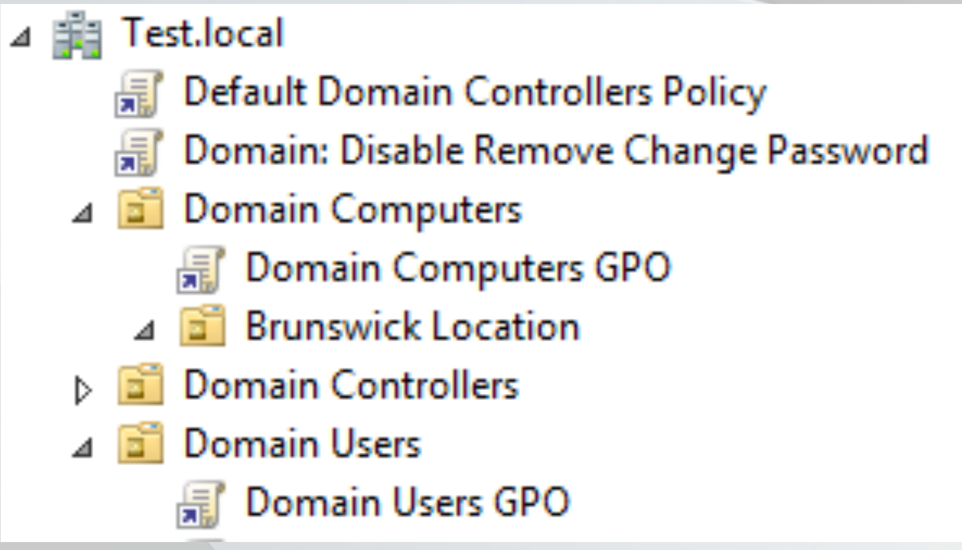
- You can change how and how often Group Policy updates
  - Settings found at Computer Config/Admin Templates/System/Logon and /Group Policy
  - Ex: Set Group Policy Refresh for computers < side benefit – desktop (background/shortcuts) will refresh
- You can change Group Policy processing method:
  - Settings found at Computer Config/Admin Templates/System/Logon and /Group Policy
  - Ex: Always Wait for the Network at Computer Startup and Logon < will slow down machines but will run synchronous

Group Policy Results

- Group Policy has two nodes: computer and user. Policies can be configured from each node and policy applies for both object types (computer/user)
- Example:
  - We have a computer that is located under the OU *Domain Computers*
  - We have a user that is located under the OU *Domain Users*

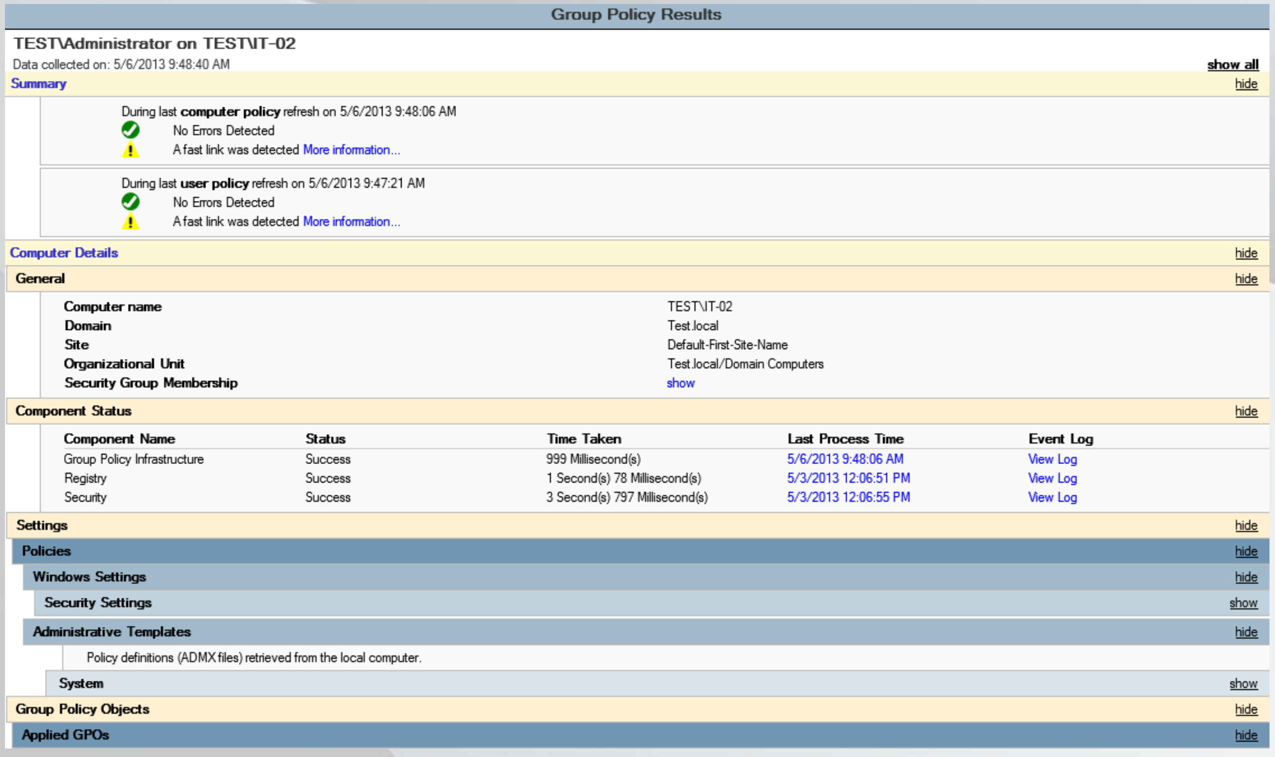
Group Policy results

- Our computer will process any GPOs applied to it (such as the domain computers GPO). It will not process the Domain Users GPO.
- Our user will process any GPOs applied to it (such as the domain users GPO). It will not process domain computers GPO.
- But how can we exactly know what our machines process?



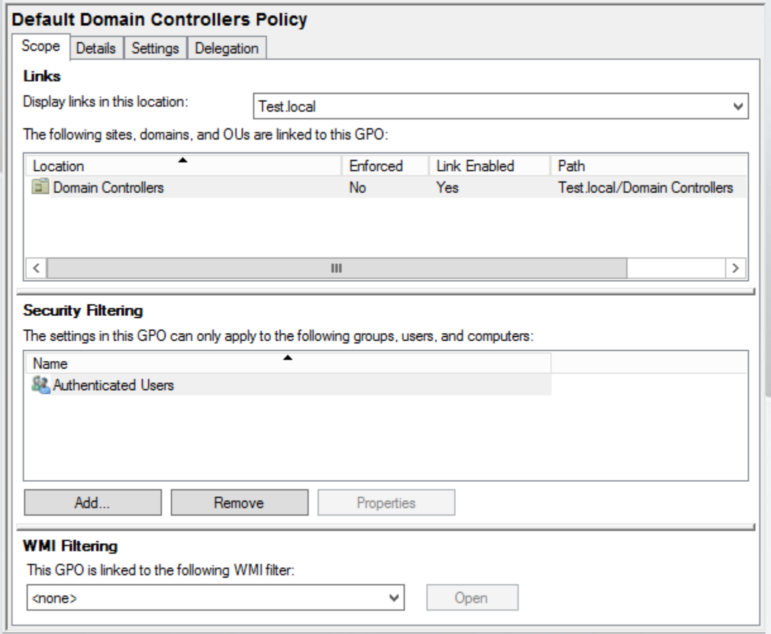
View GPresult

- Command Line: GPResult /r
- GUI: GPResult /h Report.htm



Scope Tab

- The scope tab of each GPO controls:
  - Links
  - Security Filtering
  - WMI.
- These three items control if a GPO applies – all must be true!
- By default, a newly created GPO is unlinked, applies to Authenticated Users, and doesn't have a WMI filter.



Security Filtering

- GPOs can be filtered to: Computers, Users, or Security Groups.
  - Special Groups
    - Domain Computers = All Computers
    - Domain Users = All Users
    - Authenticated Users = All Computers and Users

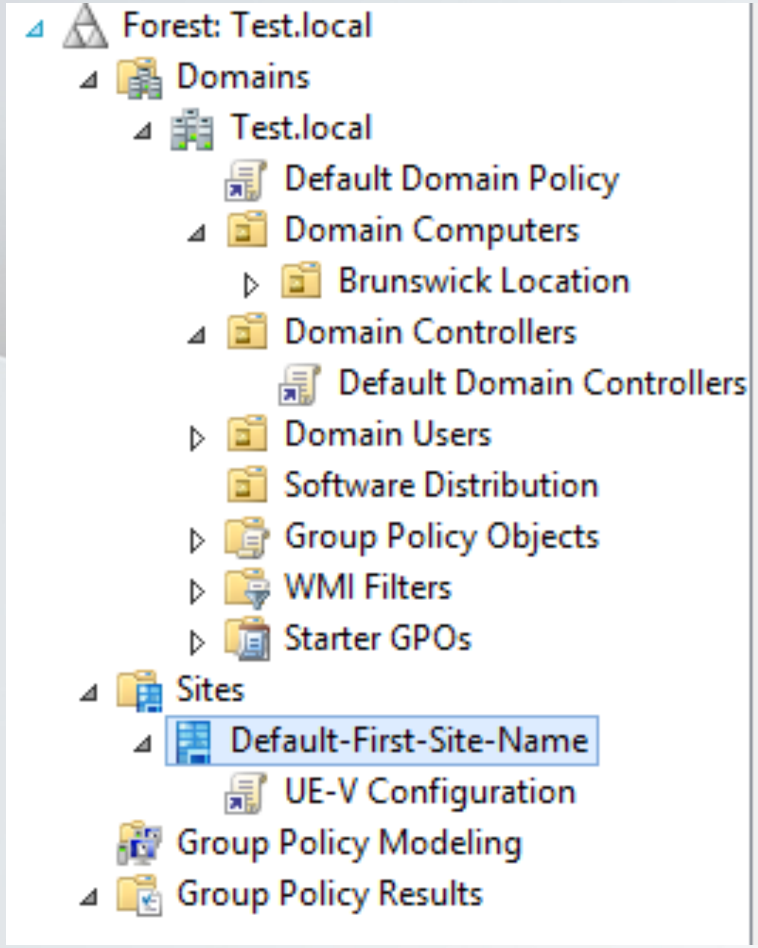
- As a best practice, GPOs should be not be filtered to specific computers or users. GPOs should be filtered to groups

Applying to Specific Objects

- If you configure a computer side setting, it must be linked to computers
- If you configure a user side setting, it must be linked to users
- You can never link a GPO to an OU containing just groups. Groups cannot process Group Policy (odd, I know…)

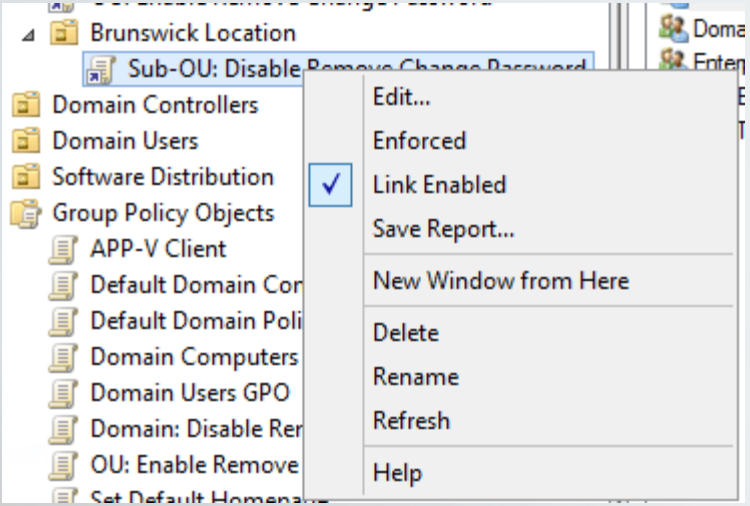
GP Linking

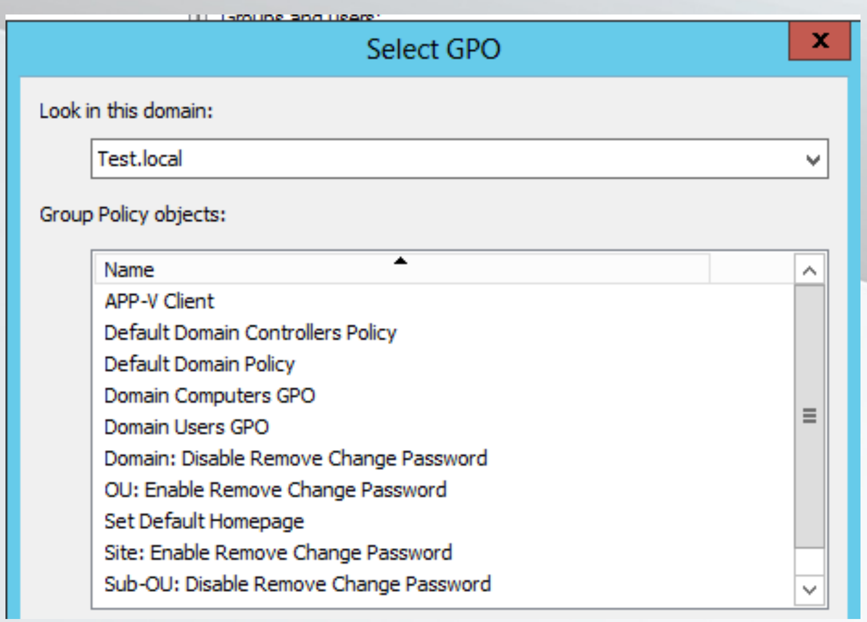
- Policies can be linked at:
  - Sites
  - Domains
  - OUs
- Policies cannot be linked at containers (such as Computers, Users, or System) or to a forest



Enabling and Disabling Links

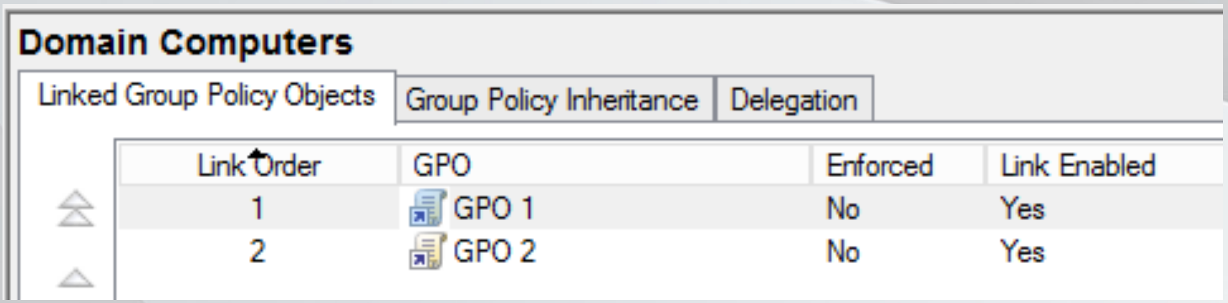
- You can *link an existing GPO* by right clicking on an OU, Domain, or Site.
- Enabling/Disabling is known as link status.
- You can unlink a GPO by right clicking on it in the OU and deselecting *Link Enabled*





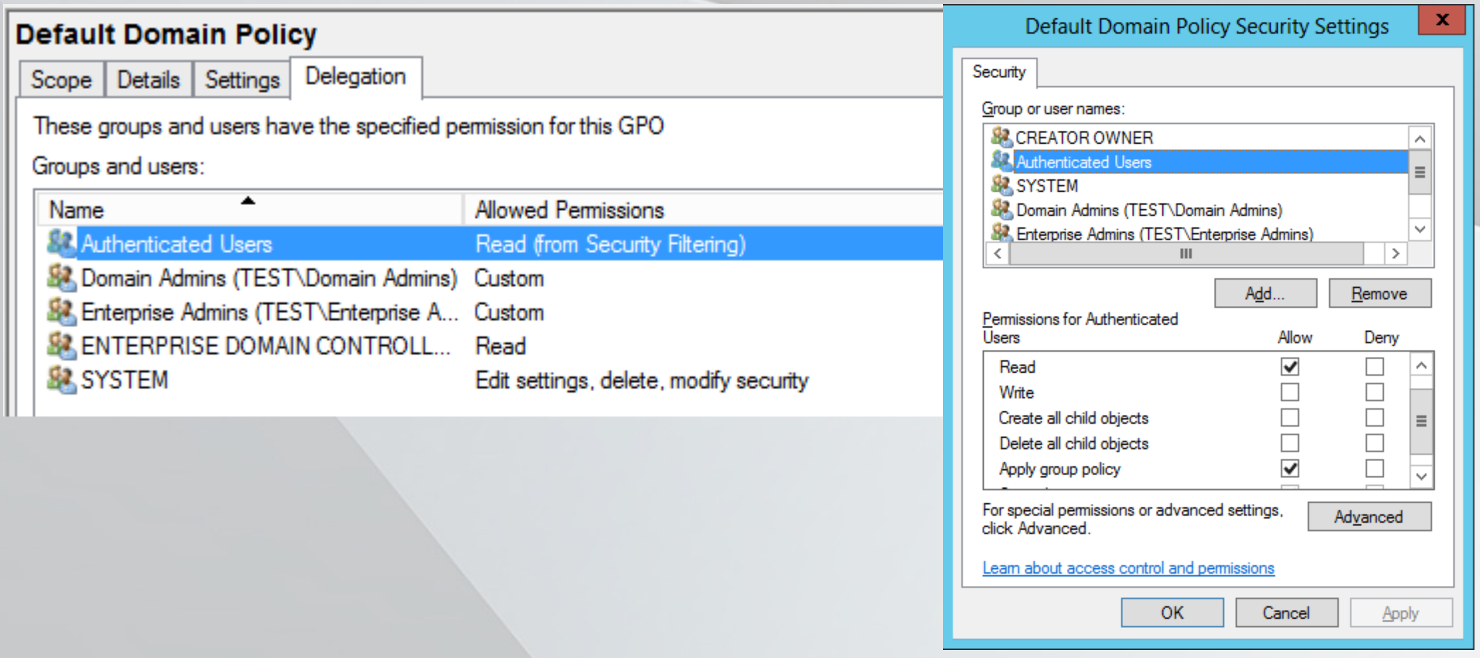
Link Order

- Multiple GPOs can be linked to a single OU.
- Last GPO linked is the first policy to process
- GPOs with a lower links order process later (1 is last)
- Can be adjusted with arrows on the left.



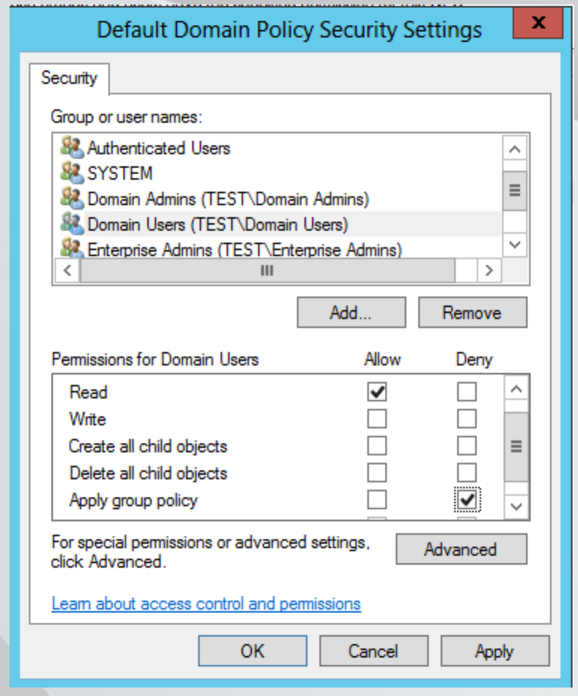
Advanced Permissions

- When an object is added to the scope tab, two permissions are added for that object: Read and Apply Group Policy
- You can see these permissions in the Delegation Tab and under Advanced.



Advanced Denying

- You have a GPO applying to Authenticated Users but want to exclude a specific group.
- Select Delegation – Advanced and Add your Group. Then Deny Apply Group Policy



LSD OU

- Like onions, Group Policy has layers (of processing).

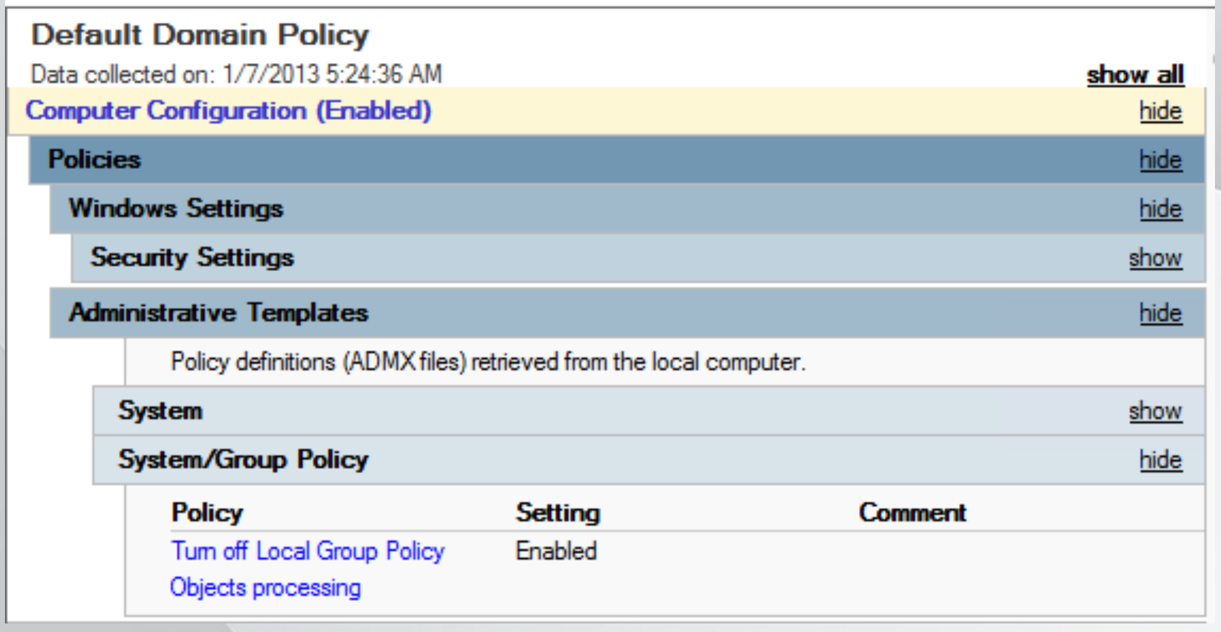
- Layer 1: Local Group Policies (First Writer)
- Layer 2: Site Linked Group Policies
- Layer 3: Domain Linked Group Policies
- Layer 4: OU linked Group Policies (Last Writer)

- Anytime in Group Policy, the last writer always wins. If you have conflicting settings (let’s say in a domain policy and an OU policy), the closest layer wins

- Sub-OUs win over higher OUs.

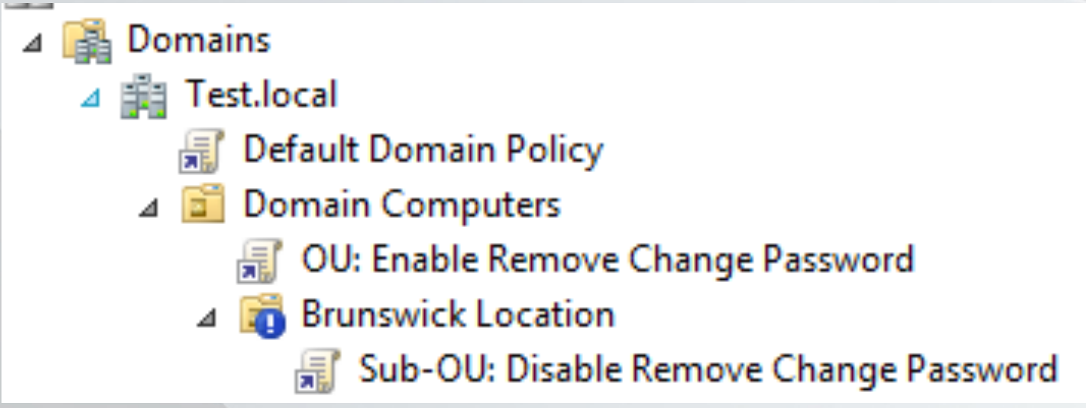
Disable Local Processing

- In an domain where Group Policy is used it makes sense to turn off local group policy.
- To do so, Enable Computer Configuration/Policies/Administrative Template/System/Group Policy/Turn Off Local Group Policy Objects processing



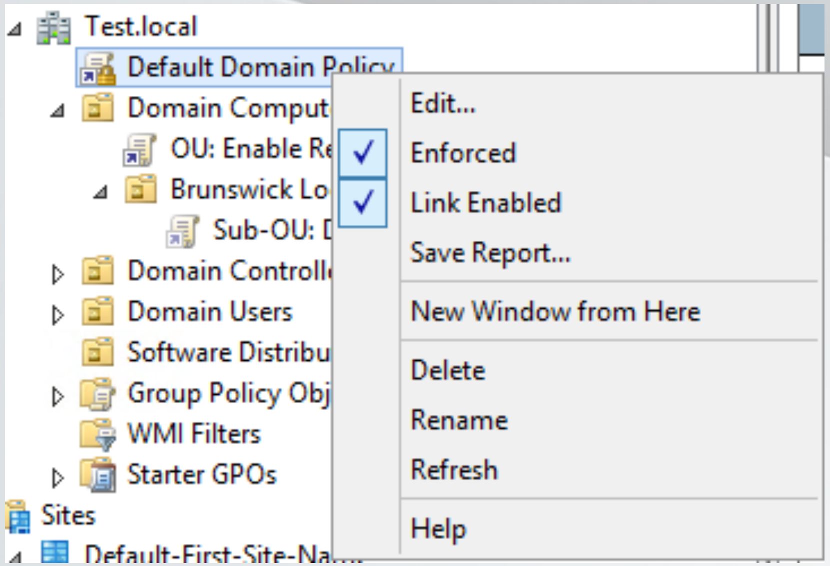
Block Inheritance

- Block Inheritance allows you to cut out any GPO from above.
- If set on an OU, domain GPOs and higher OU GPOs will not apply.
- Why use block inheritance when lower GPOs win anyways?
  - Remember that lower only wins if a setting contradicts a higher setting. Block Inheritance blocks all setting from processing (without you configuring a GPO)



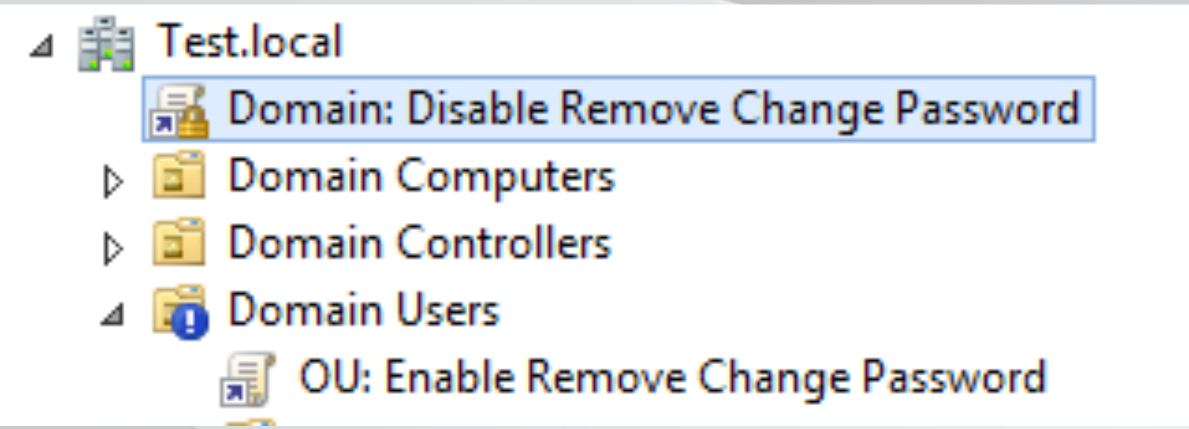
Enforcement – LSD OU upside down

- Enforcement allows you to set GPOs from on high without being overwritten.
- To enforce, right click on a GPO link and select enforce. This setting is per link – not per GPO.
- Enforced GPOs apply backwards
- Use sparingly or not at all



Enforcement and Block Inheritance

- You have an enforced GPO linked at the domain
- You have block inheritance on an OU.
- The enforced GPO will win. Enforced ignores block inheritance



Loopback processing

- Remember these rules?
  - If you configure a computer side setting, it must be linked to computers
  - If you configure a user side setting, it must be linked to users

- But what if you want to apply a user side setting (ex: a home page) to a group of computers? Use Loopback!

- Loopback can be configured at *Computer Configuration/Administrative Templates/System/Group Policy/Configure user Group Policy loopback processing mode.*

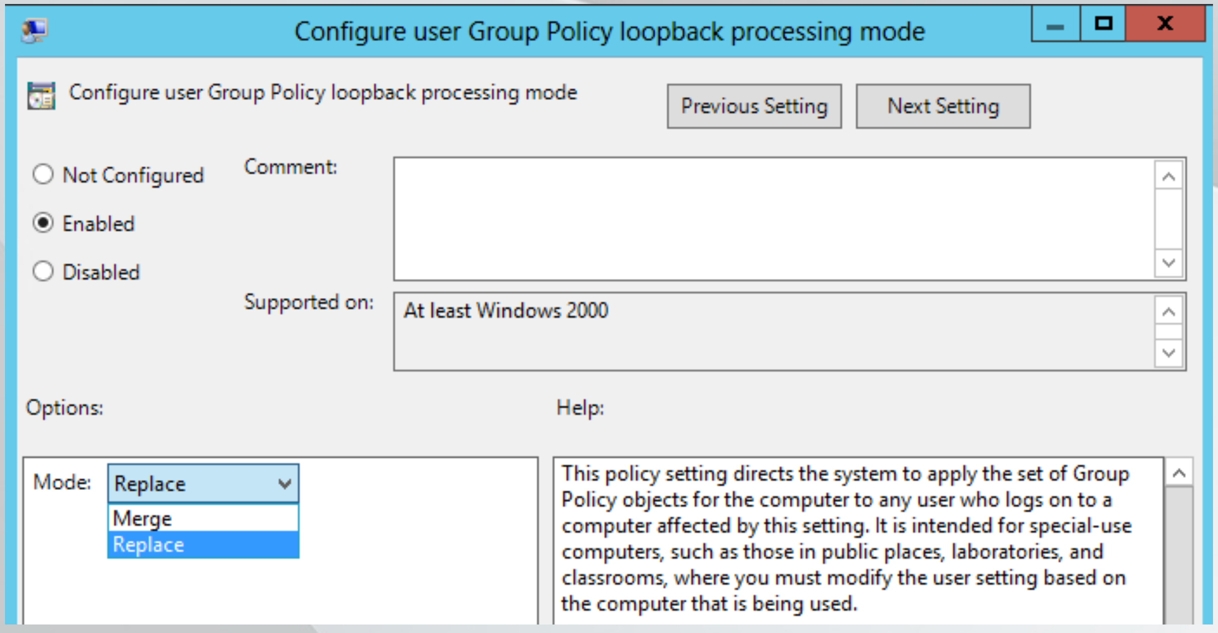
How does Loopback work

- No Loopback
  - Computer Starts/Contacts DC
  - Gets all GPOs linked to it
  - Processes Computer Node (ignores user node)
  - Presents Login Screen
  - User Authenticates
  - Gets all GPOs linked to it
  - Processes User Node (ignores computer node)
  - User is logged in

- With Loopback
  - Computer Starts/Contacts DC
  - Gets all GPOs linked to it
  - Processes Computer Node (ignores user node)
  - Presents Login Screen
  - User Authenticates
  - Gets all GPOs linked to it
  - Processes User Node (ignores computer node)
  - Computer says “shoot, I can be a user too”
  - Gets all GPOs linked to it (the computer)
  - Processes User Node (ignores computer node)
  - User is Logged in

Additional Info about Loopback

- Comes in two modes: Merge and Replace
  - Merge adds in the user's setting + user node settings linked to the computer
  - Replace only uses user node settings linked to computer
- Loopback changes processing for the entire computer – not just one GPO.



Loopback Modes: Merge and Replace

- Merge Mode
  - Computer Starts/Contacts DC
  - Gets all GPOs linked to it
  - Processes Computer Node (ignores user node)
  - Presents Login Screen

  - User Authenticates
  - Gets all GPOs linked to it
  - Processes User Node (ignores computer node)

  - Computer gets all GPOs linked to it
  - Processes User Node (ignores computer node)
  - Computer merges it's extra settings to the User's RSOP
  - User is Logged in

- Replace Mode
  - Computer Starts/Contacts DC
  - Gets all GPOs linked to it
  - Processes Computer Node (ignores user node)
  - Presents Login Screen

  - ~~User Authenticates~~
  - ~~Gets all GPOs linked to it~~
  - ~~Processes User Node (ignores computer node)~~

  - Computer gets all GPOs linked to it
  - Processes User Node (ignores computer node)
  - Computer replaces all settings grabbed by the user with it's User Node settings

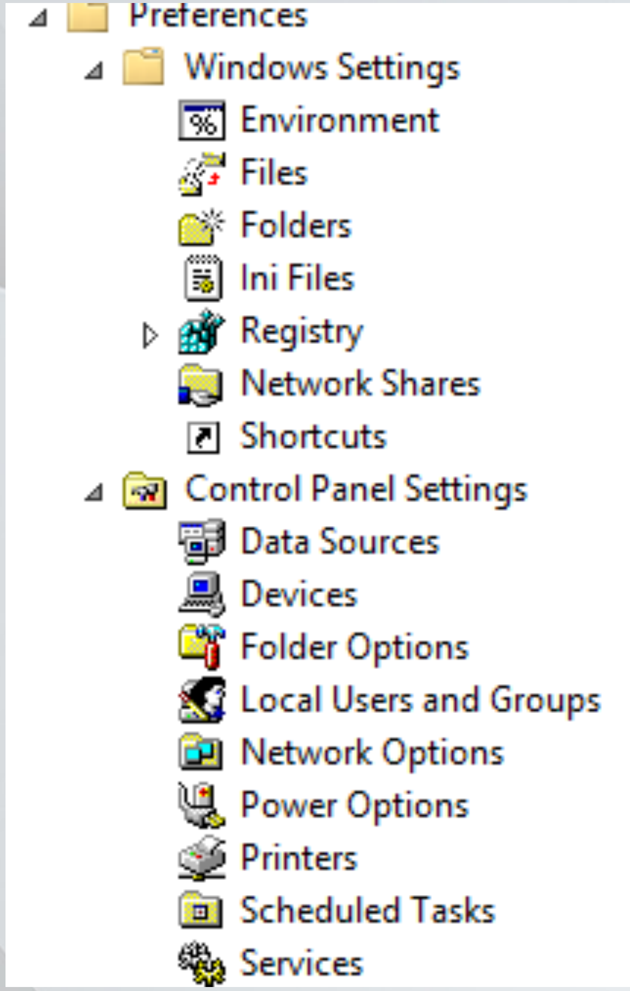
Policy/Policy Setting/Registry Policy/ADMX

- All of these names mean the same thing –everything under Administrative Templates. For simplicity, we will try to call them Policy
- Stored in C:\windows\PolicyDefinitions\ on every single machine
- Language independent and XML based.
- Available for 3rd party software: Office, Adobe, NetSupport, etc

The Two Traits of Policies

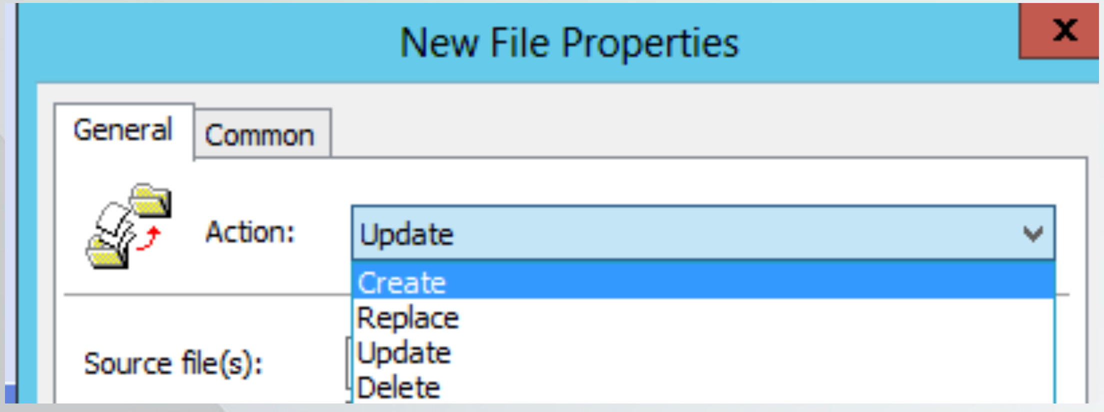
- Not Tattoo Settings
  - If you configure a setting (ex: User Config/Administrative Templates/Windows Components/Internet Explorer/Disable changing home page settings) and later remove that setting from the GPO, the computer will also remove setting.
- Takes Precedence
  - A policy will always be the default setting once configured. When you configure a specific setting, the user will not be able to override the setting.

Group Policy Preferences



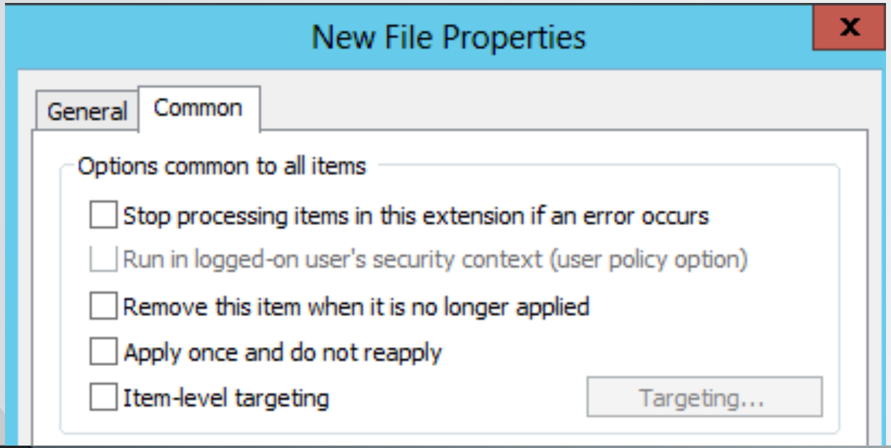
CRUD! 4 More Things to Remember

- Create: creates the preference item. Will not edit anything that already exists.
- Replace: deletes the existing item and creates a new one. Will not create if item doesn't exist.
- Update: will create item if it doesn't exist. Replace if it does.
- Delete: well, deletes the item
- Preferences without CRUD default to Update (ex: Internet Explorer Properties)



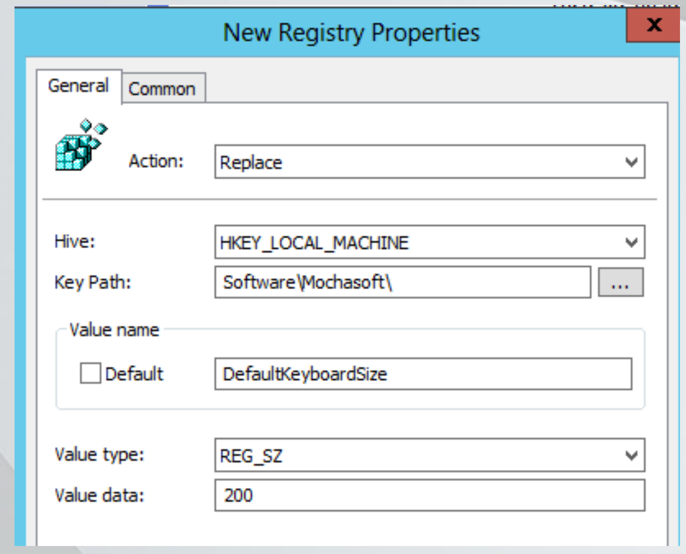
Common Tab

- Stop Processing: Used to speed up GPPs instead of letting a GPP error out
- Run in: Used for least privilege as GPPs almost always run in system context (Drive Mappings/User Printers do not)
- Remove: We'll talk about this one:
- Apply Once: It is serious about what it says – will not even apply again with a GPUpdate /force
- ILT: WMI on Steroids



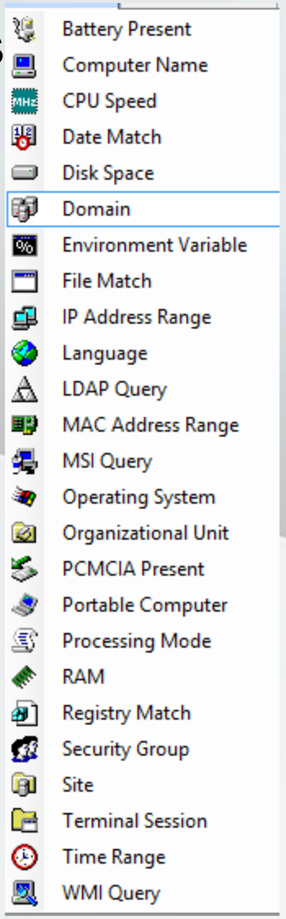
Remove this Item When it No Longer Applies

- When checked, the preference item is switched to replace.
- When preference falls out of scope, the entire item is removed (not just the specific setting)
- DefaultKeyboardSize registry key isn't set back to 100 or even to a blank setting – the entire item is deleted and the application breaks



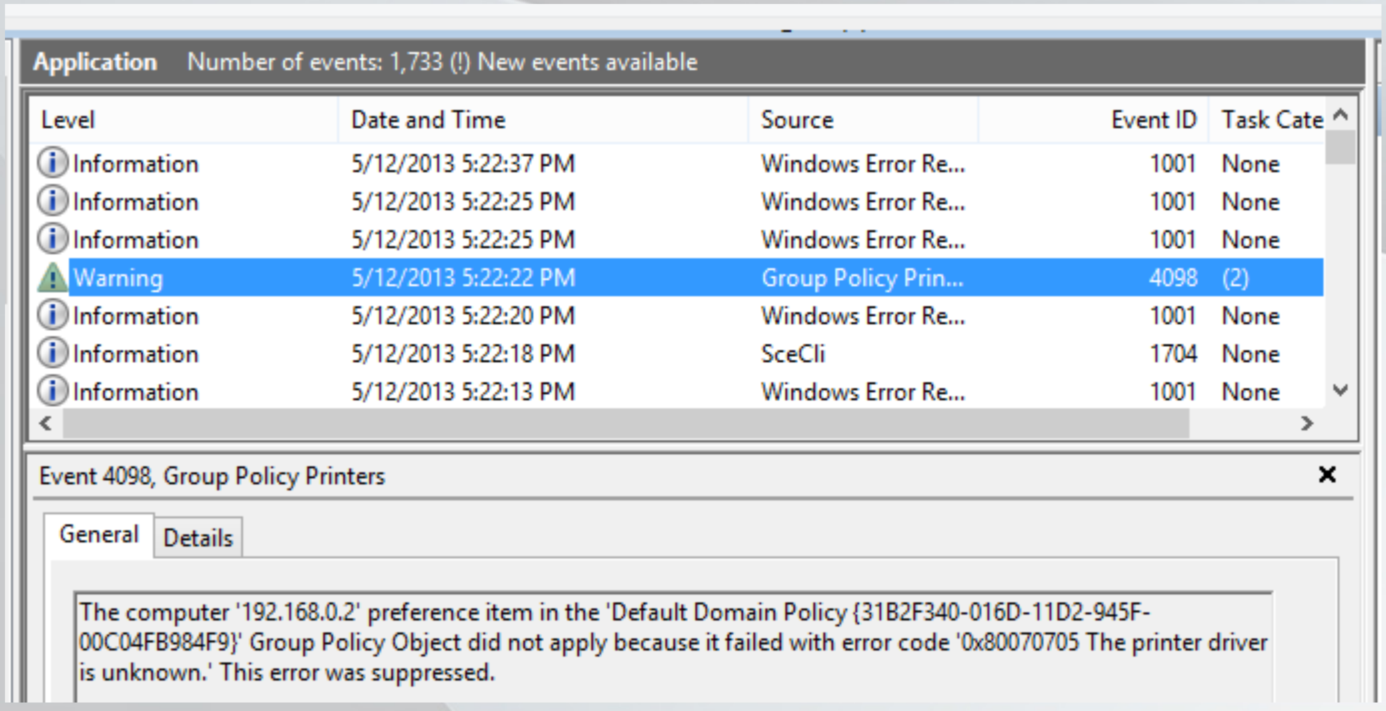
ILT

- Item Level Targeting is a GUI based filtering that acts like WMI (ex: computer must match this name or be a member of this OU)
- Multiple targets can be combined with AND or OR statements
- Can use NOT statements



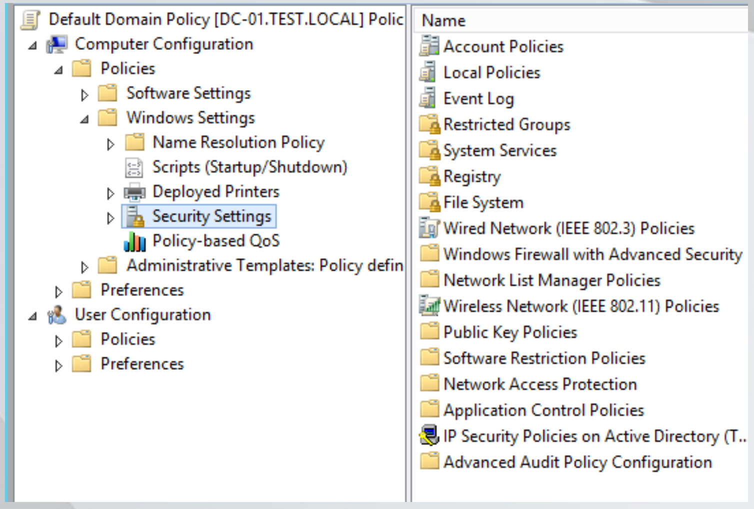
Troubleshooting Preferences

- First place to start: Application in the Event Viewer
- Second place: Group Policy detailed log (Event Viewer\Applications and Services Log\Microsoft\Windows\Group Policy



Security

- Found under Computer Configuration\Windows Settings\Security Settings
- Manage rights, auditing, local machine settings
- Manage Group Membership
- Set File/Registry Permissions
- Configure Firewall, Software Restriction/AppLocker
- Set Wireless Profiles



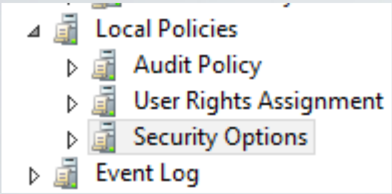
Security Processing

- When will a GPUpdate happen?
- How often does it happen?
- Will a policy reapply if it hasn’t been changed?

- The Security CSE will automatically reprocess all settings every 16 hours regardless of changes in the GPO.

Security Settings: Local Policies

- Not the same a local GPO – these are security settings
- Three sections: Audit, User Rights Assignment, Security Options
  - User Rights Assignment control permissions such as Deny Log On Locally and Run as a Batch
  - Security Options control device restrictions, Interactive Logon (CTRL+ALT+DEL), UAC, etc



SCRIPTING

Common characteristics of scripting languages

- Interpreted vs compiled
- Text based vs binary
- Native vs cross-platform

WINDOWS BATCH FILES

What is a windows batch file

- A batch file is an unformatted text file or script file which contains multiple commands to achieve a certain task. It contains series of commands that is executed by **command line interpreter**.
- The instructions in batch files are for automating repetitive command sequences.

Editing batch files

- Windows batch files are easily readable by any text editor- one of the most popular one notepad.exe (even in CORE edition)
- Name command prompt editors !
- **Exercise 1** Create new file from command line quick exercise

Batch file extensions

- Traditional extension .bat
- Modern extension .cmd

- Biggest difference in error handling

Commenting in batch

- **Documenting code with comments**
  This is one of the most important one because without proper documentation it becomes tedious to maintain the code and debug it

  So it is always a good idea to insert comments in programs or codes, explaining what the next lines or block of code is trying to accomplish, how and why.
  Either REM or : : is used for comments in batch file programming.

```
REM This is a comment
:: This is also a comment
```

Executing windows batch file

- Typically double clicking should be enough
  - Terminal
  - Full path vs local directory
  - compspec
    - Child processes
    - Auto-run scripts

@echo off

- @ is used to hide the line from printing on the command prompt
- @echo off turns off printing of the commands for the whole script execution or until echo is turned on again

Batch commands

- As mentioned batch commands are built into the command interpreter
- batch commands are NOT case sensitive
  - Windows overall is not case sensitive
- Most important command – HELP
  - Will display all available commands available in current command interpreter
  - Help [command] will return help page for the command
  - Typical alias for help is /?

Exercise 2

- Create new file FILE1.TXT in command prompt
- Create folder FOLDER1
- Copy the file FILE1.TXT to the folder FOLDER1
- Rename the file FILE1.TXT to FILE2.TXT
- Rename the folder FOLDER1 to FOLDER2
- Clear command prompt window
- Create folder FOLDERX
- Copy the folder FOLDER2 with all the files in there to the folder FOLDERX

Batch variables

- Defining variable in batch is straight forward:
  SET variable1=value1
- Variables can be either local or global
  @echo oFF
  SET var1=globalvariable
  SETLOCAL
  SET var2=localvariable
  ECHO %var1%
  ENDLOCAL
  ECHO %var2%
  PAUSE
- Checking if variable is used:
  IF DEFINED var1 (ECHO var1 IS defined) ELSE (ECHO var1 is NOT defined)

Batch functions

- Function defining is as following
  :function1
  different operations
  EXIT /B 0
- Where /B 0 is to ensure successful termination or proper exit of the functions
- To call functions CALL is used
  CALL function1 or CALL function2 param1 param2
  @echo OFF
  CALL :param_function value1, value2
  EXIT /B %ERRORLEVEL%
  :param_function
  ECHO The values are %~1 is %~2
  PAUSE
  EXIT /B 0

Batch functions with return values

- Occasionally it is necessary to use the values returned from a function

```
@echo OFF
CALL :retun_value_function ret_val1,ret_val2
ECHO The square root of %ret_val1% is %ret_val2%
PAUSE
EXIT /B %ERRORLEVEL%
:return_value_function
SET %~1=100
SET %~2=10
EXIT /B 0
```

Batch If Else statements

- SET /A a=2
  SET /A b=3
  SET name1=Aston
  SET name2=Martin
  :: Using if statement
  IF %a%==2 echo The value of a is 2
  IF %name2%==Martin
  echo Hi this is Martin
  :: Using if else statements
  IF %a%==%b% (echo Numbers are equal) ELSE (echo Numbers differ)
  IF %name1%==%name2% (echo Name is Same) ELSE (echo Names differ)
  PAUSE

Batch loops

- Syntax for executing FOR loop is as follows
  FOR %%var_name IN list DO example_code
- Important to notice is that the variable is defined %%var_name instead of %var_name%

  ```
  @echo off
  FOR %%var1 IN (a b c) DO
  ECHO %%var1
  PAUSE
  ```
- FOR /L %%var_name IN (Lowerlimit, Increment, Upperlimit) Do some_code

Batch loops in files and folders

- In file and folder looping %var is used instead of %%var
  FOR %y IN (C:\Windows*) DO @ECHO %y

  FOR /D %y IN (C:\Windows*) DO @ECHO %y

WINDOWS SCRIPTING HOST

Windows Scripting Host

- WSH
  - VBScript
  - Jscript
- Other scripting hosts
  - Internet Explorer
  - Exchange
  - IIS
- ActiveX scripting language

Wscript vs Cscript

Exercise 3

- First create an example file justtext.vbs
  - WScript.Echo “Just some text in box”
- Secondly create an example with color asking – color.vbs
  - Wscript.echo “getting started”
    askColor = InputBox(“What is your favourite color?”)
    MsgBox askColor & “ is a very beautiful color !”
- Run the scripts both with wscript and script name
- Now run them with cscript

Script writing notepad vs IDE

- Notepad
  vs notepad++
  vs IDE / ISE

VBScript functions

- Functions to simplify scripting and take advantage of the code reuse
- Built in functions
  - Date()
- Own functions
  - IsMember()
- **Exercise 4**
  Create a function to check if user (ask for user with pop up box) is a member of a group (administrators” and then return also the result

VBScript Subroutines

- Differences in subroutines vs functions

```
 Sub MapDrive(sLetter, sUNC)
      Set oNet = _
      WScript.CreateObject(“WScript.Network”) oNet.MapNetworkDrive sLetter, sUNC
End Sub
Sub MapPrinter(sUNC)
      Set oNet = WScript.CreateObject(“WScript.Network”)       oNet.AddWindowsPrinterConnection sUNC oNet.SetDefaultPrinter
sUNC
End Sub
```

WINDOWS MANAGEMENT INSTRUMENTATION

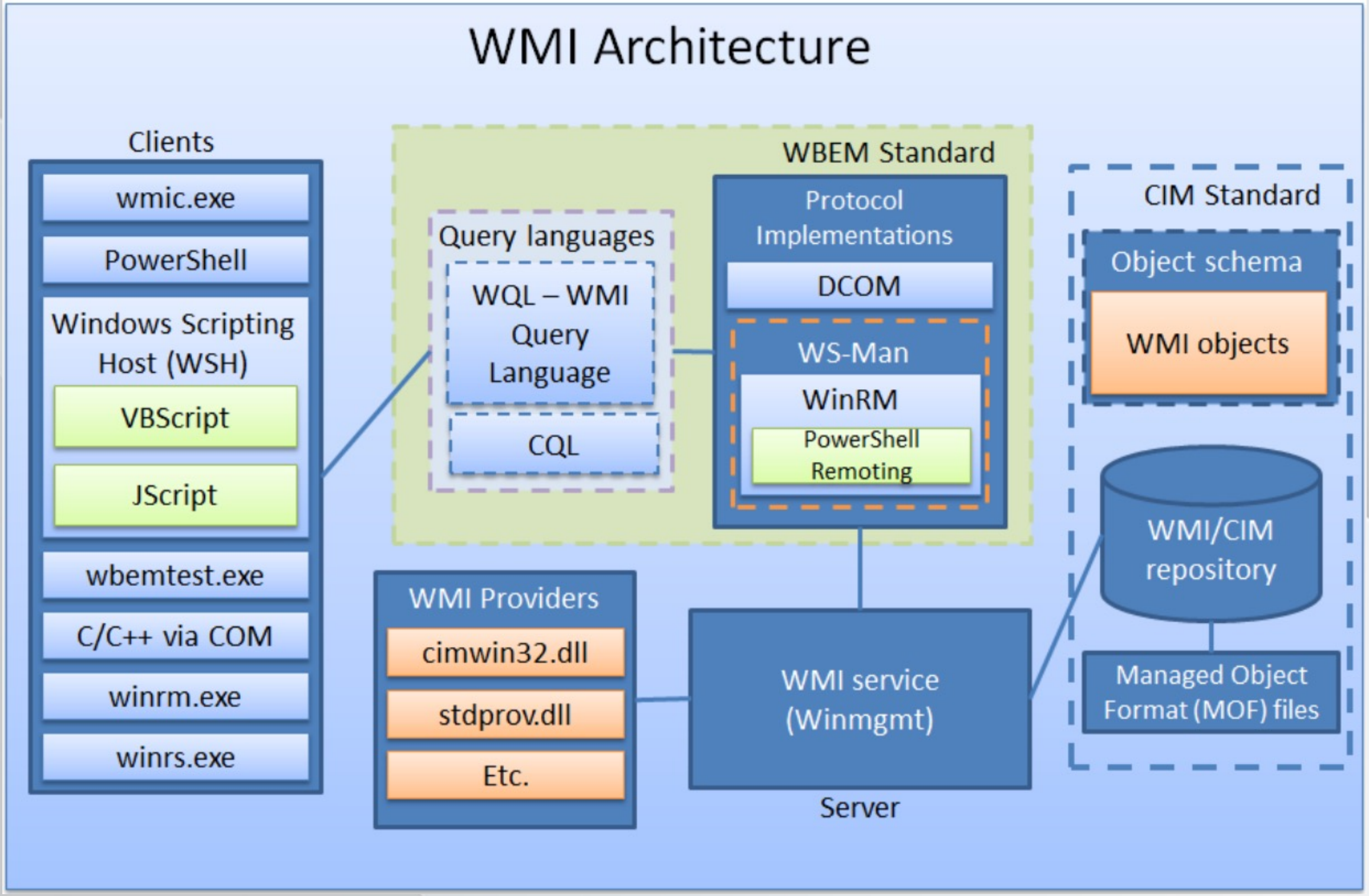
What is WMI

- WMI is the Microsoft implementation of the Web-Based Enterprise Management (WBEM) and Common Information Model (CIM) standards published by the Distributed Management Task Force (DMTF). Both standards aim to provide an industry-agnostic means of collecting and transmitting information related to any managed component in an enterprise. An example of a managed component in WMI would be a running process, registry key, installed service, file information, etc. These standards communicate the means by which implementers should query, populate, structure, transmit, perform actions on, and consume data

WMI structure

- The WMI infrastructure consists of four components: WMI providers, the Common Information Model Object Manager (CIMOM), the CIM repository, and the WMI scripting library.
  - WMI providers, many of which are built into the Windows OS and called standard providers, ensure that managed resources are available and addressed appropriately in WMI.
  - CIMOM acts as a traffic director that handles interactions between WMI providers and consumers.
  - The CIM repository is a schema that models and manages every piece of data available through WMI. The data is categorized into classes, and the classes are logically grouped into namespaces. Most computer- and OS-related information resides in the root\cimv2 namespace.
  - The WMI scripting library is the interface through which scripting tools such as JScript, VBScript, and other languages (e.g., Perl) that support Microsoft ActiveScripting technology easily consume WMI data.

WMI archidecture



Run a program through WMI

- VB
  Const HIDDEN_WINDOW = 0
  strComputer = ".”
  Set objWMIService = GetObject("winmgmts:" & "{impersonationLevel=impersonate}!\\" & strComputer & "\root\cimv2")
  Set objStartup = objWMIService.Get("Win32_ProcessStartup")
  Set objConfig = objStartup.SpawnInstance_ objConfig.ShowWindow = HIDDEN_WINDOW
  Set objProcess = GetObject("winmgmts:root\cimv2:Win32_Process") errReturn = objProcess.Create("Notepad.exe", null, objConfig, intProcessID)
- PowerShell
  $startup=[wmiclass]"Win32_ProcessStartup" $startup.Properties['ShowWindow'].value=$False ([wmiclass]"win32_Process").create('notepad.exe','C:\',$Startup)
- Wmic
  wmic process call create "c:\windows\notepad.exe"

discussions